



ColorCloud

HAMBURG

24. - 25. APRIL 2025

Power Platform for the Perspective from a Business Central Developer



Tine & waldo



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#ColorCloudSpeaker 



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#ColorCloudSpeaker 

Some challenges in the world of software development

Repeatability

- Build Reusable Assets
- One-Time Effort, Many Sales
- Scalable Delivery
 - Fast, efficient
- Maintainability
 - Automated Upgrades
 - Hotfixes
- Standardized foundation
 - BC & PP Platform
- Maximize ROI



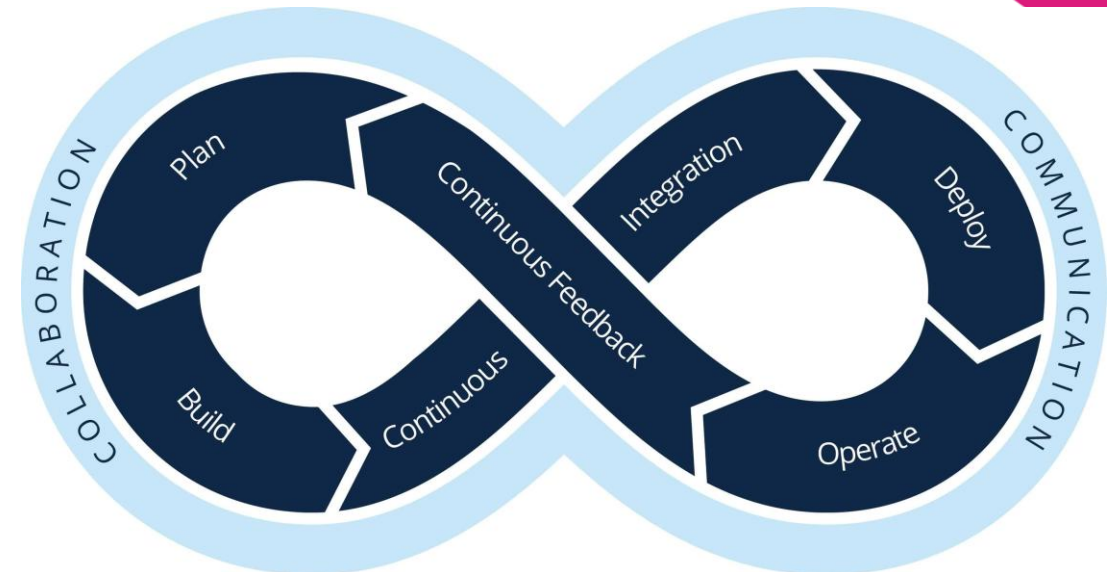
Maintainability

- Automation
 - CI/CD
 - Validation
 - Upgrades
- Environment Accessibility (GDAP)
- Product feedback
 - Telemetry



Development Methodology

- Dev environments
 - DEV / QA / UAT / PROD / TRAINING
- Versioning & branching
- VNext
 - For the product
 - For your apps



Customization

Customers are never 100% happy with an out-of-the-box product.

- They have "unique" processes
- Improve efficiency
- Meet their industry needs
- Enhance user experience
- ...

Business Central in a nutshell

What is Business Central?

Microsoft Dynamics 365 Business Central is a **cloud-based, all-in-one ERP** (Enterprise Resource Planning) solution designed for **small to medium-sized businesses**

Its **purpose** is to streamline operations—finance, sales, inventory, purchasing, and more—in a single platform.

It is built on **modular** architecture and is highly **customizable** and **extensible**.

Technical Backbone

Built on AL

Extensibility:

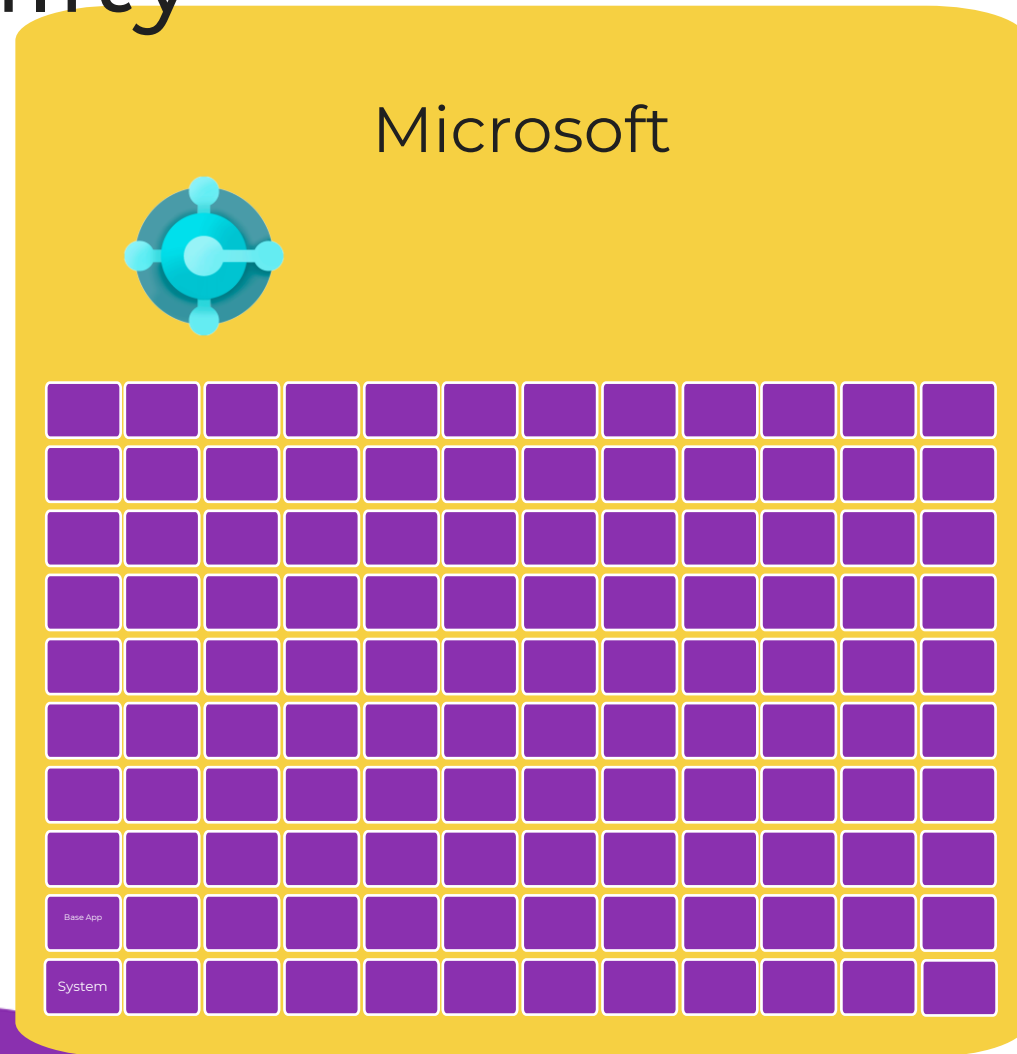
- The ability to create apps in AL
- Microsoft
 - o creates base apps (in AL)
 - o provides many "events" to be able to tap into from other apps
- Partners
 - o Create apps as products (AppSource) or specific for customers (PTE)

Upgrade-friendly:

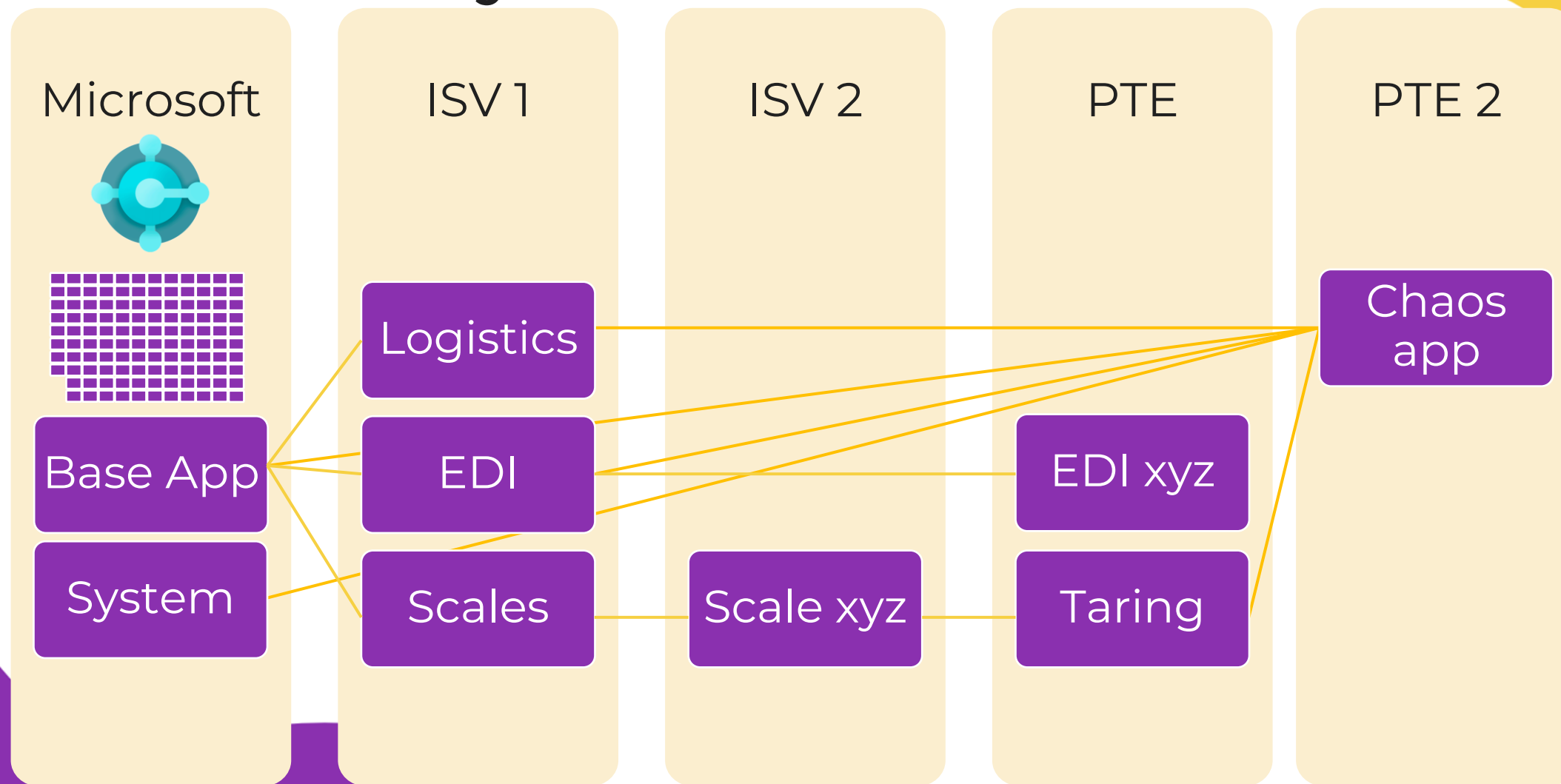
- Architecture ensures continuous updates with minimal disruption
- Extensions keep customizations separate from core code



Extensibility



Extensibility



Development Methodology

Business Central has its own language (AL), so it comes with all the benefits:

- Source control
- Branching strategies
- CI/CD
- Automated tests
- ..

Power Platform

Disclaimer

We could be completely wrong – but this is the world of PP from the eyes of a BC developer..

What is the Power Platform?

*“A **low-code platform** to build **apps**, automate **workflows**, analyze **data**, and **integrate systems**”*

What is the Power Platform?

Microsoft Power Platform

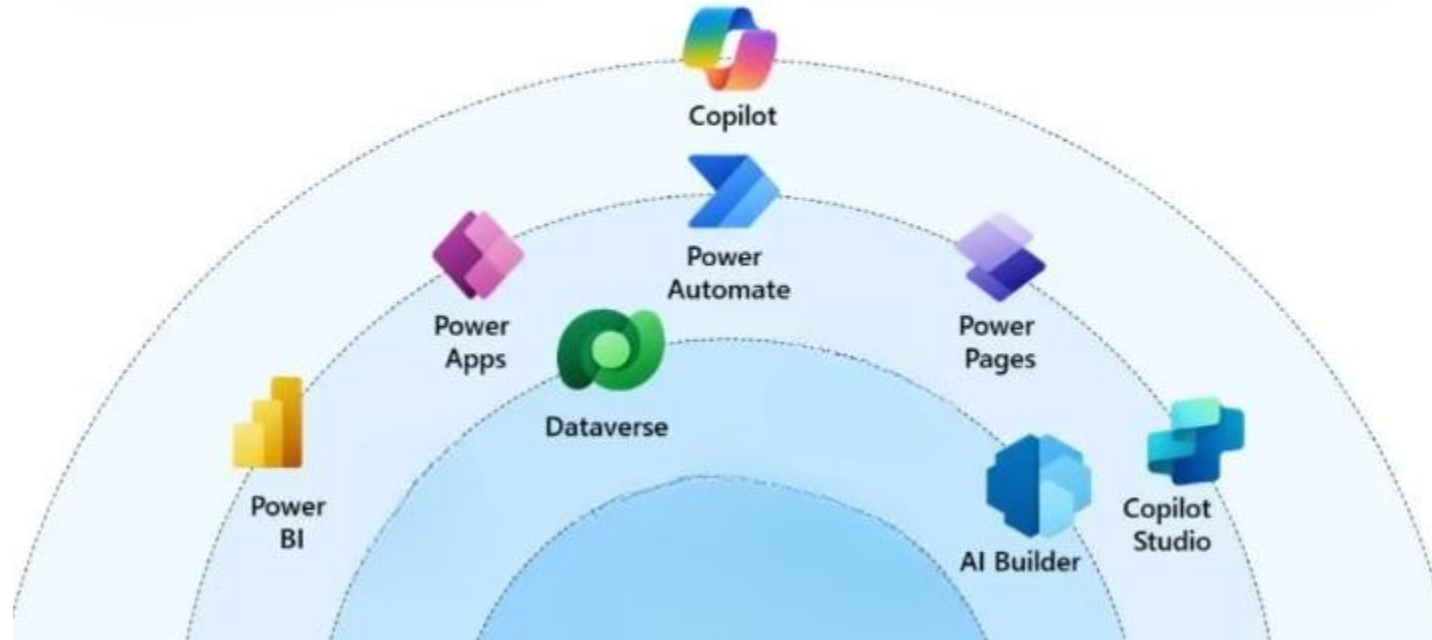
The world's **most complete** platform for developer productivity, spanning no-code to pro code



What is the Power Platform?

Microsoft Power Platform

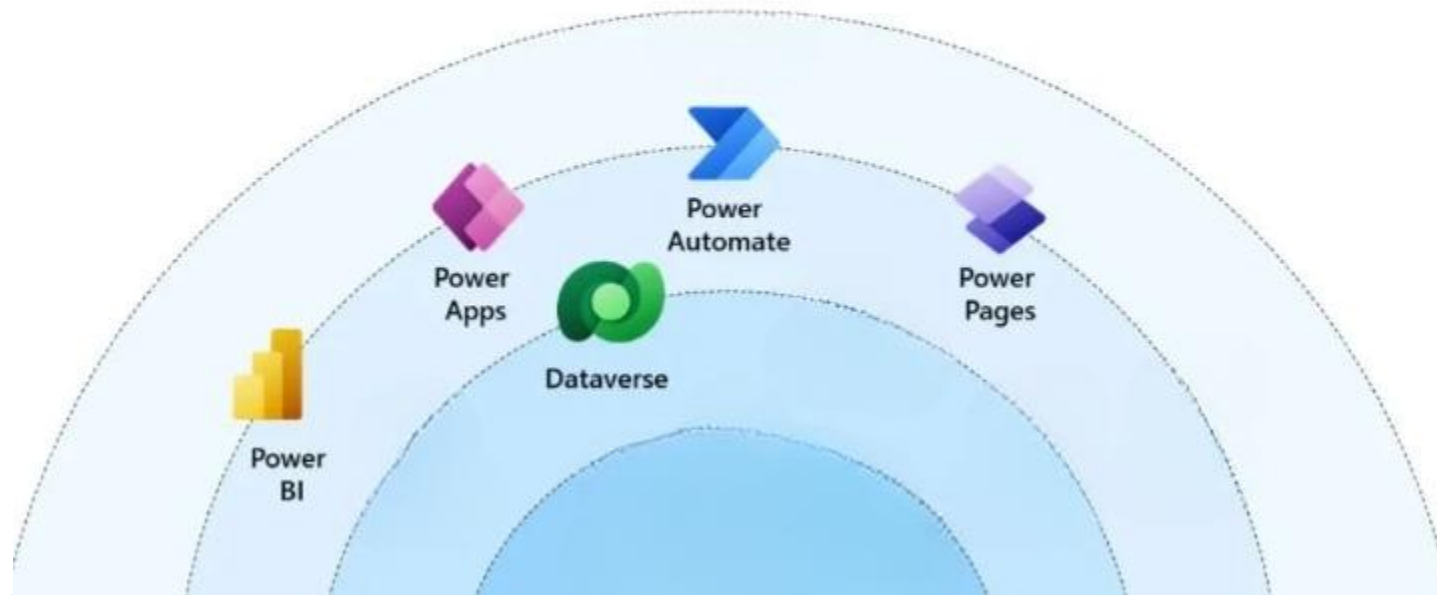
The world's **most complete** platform for developer productivity, spanning no-code to pro code



What is the Power Platform?

Microsoft Power Platform

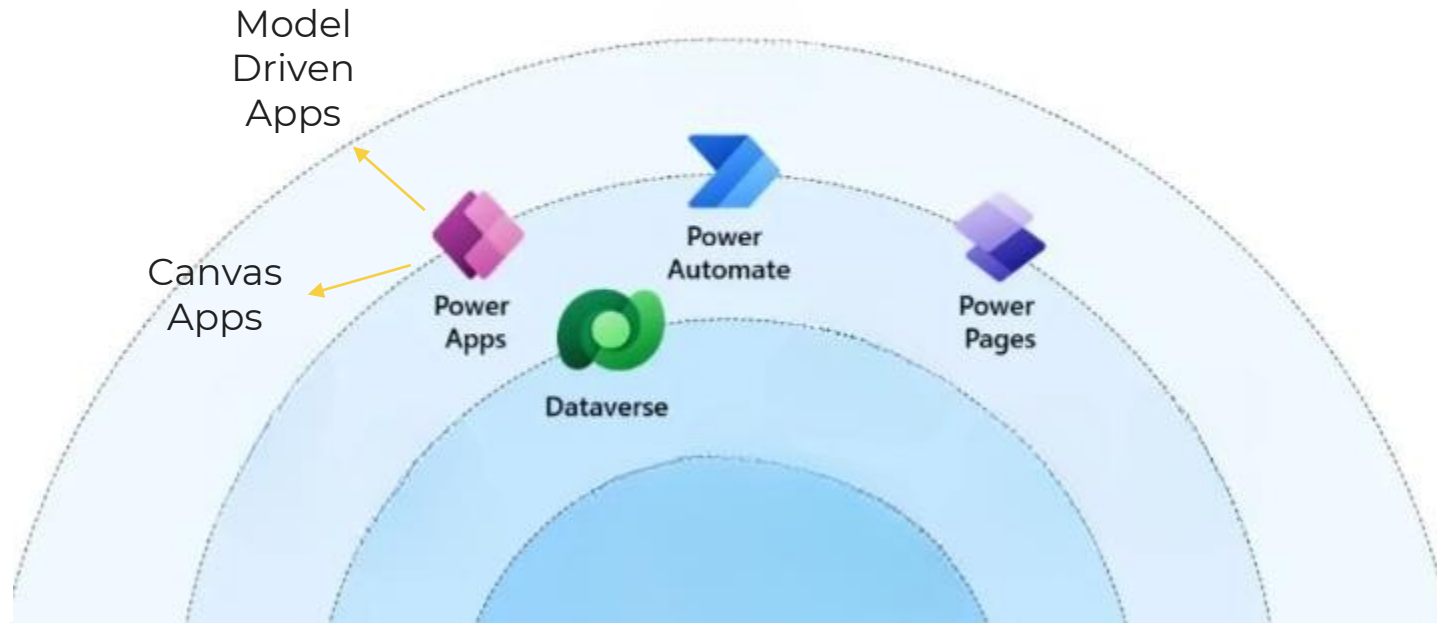
The world's **most complete** platform for developer productivity, spanning no-code to pro code



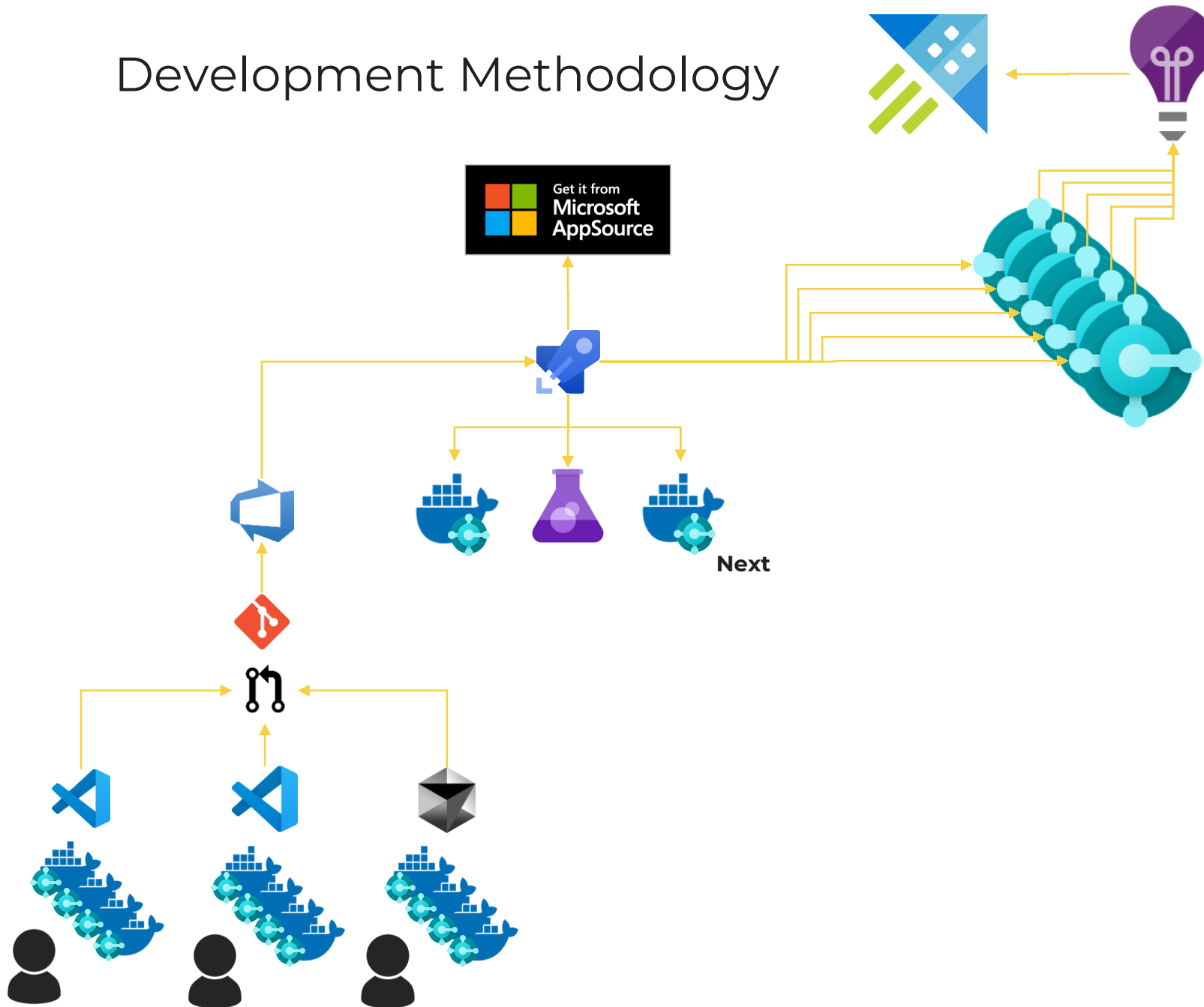
What is the Power Platform?

Microsoft Power Platform

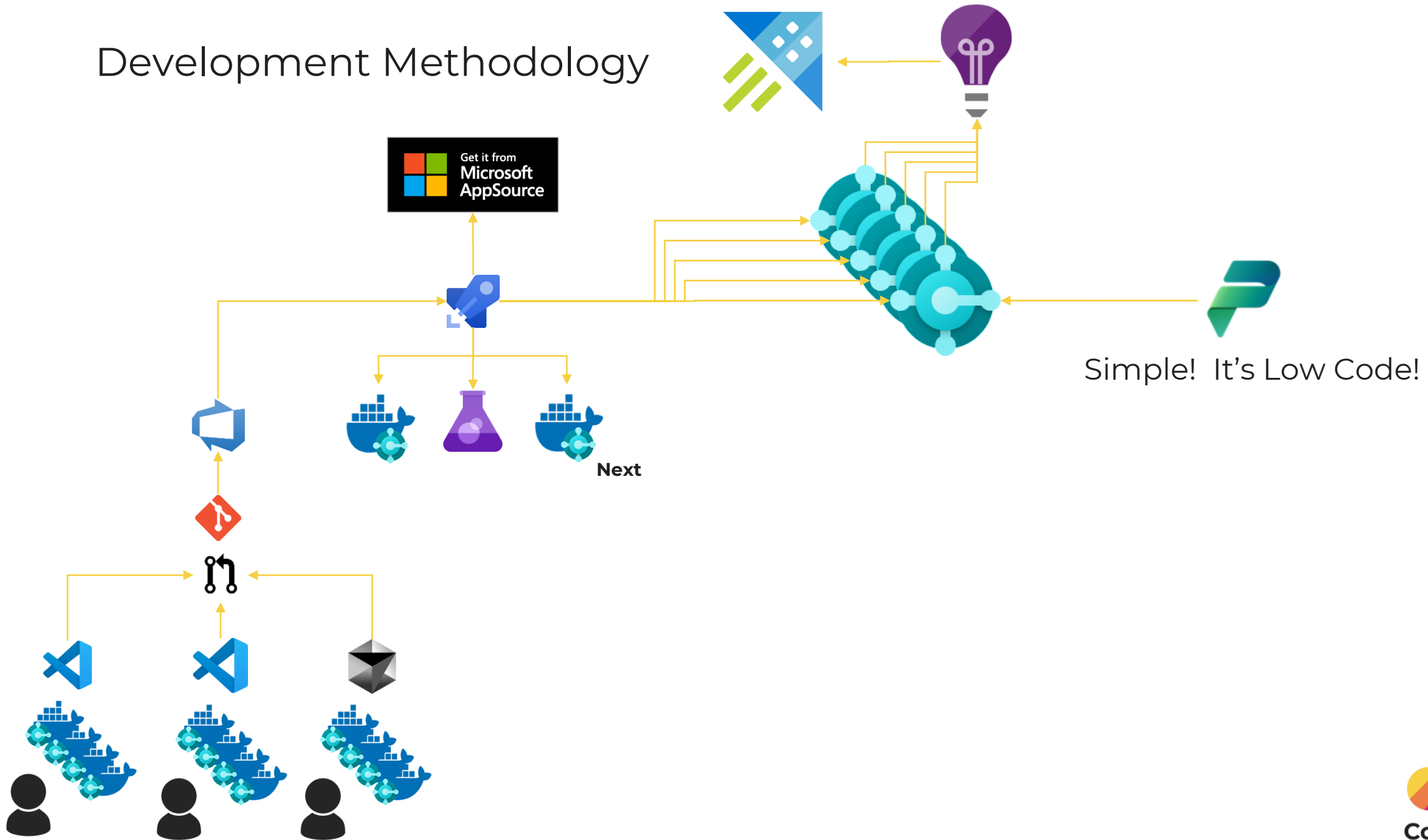
The world's **most complete** platform for developer productivity, spanning no-code to pro code



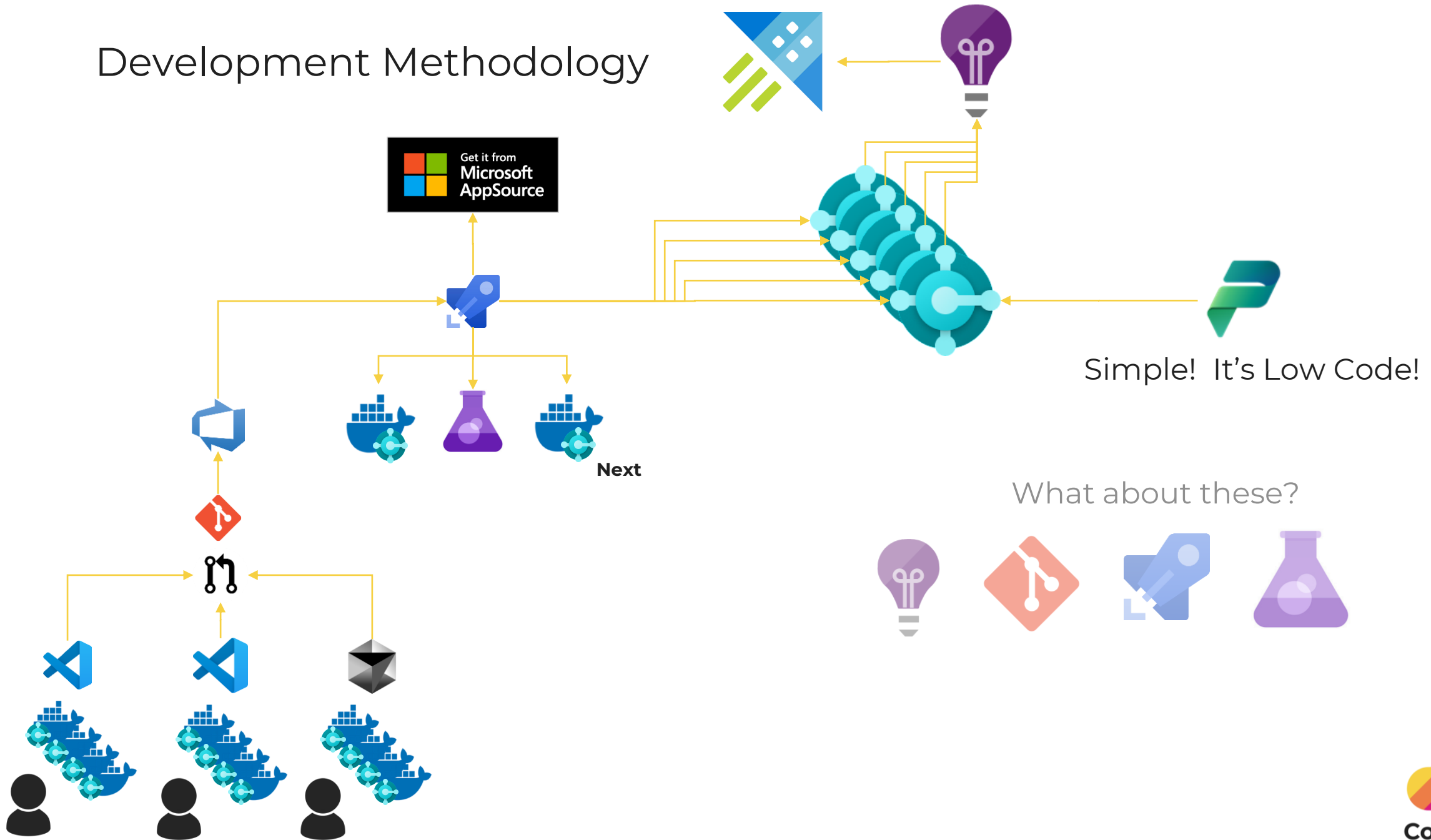
Development Methodology



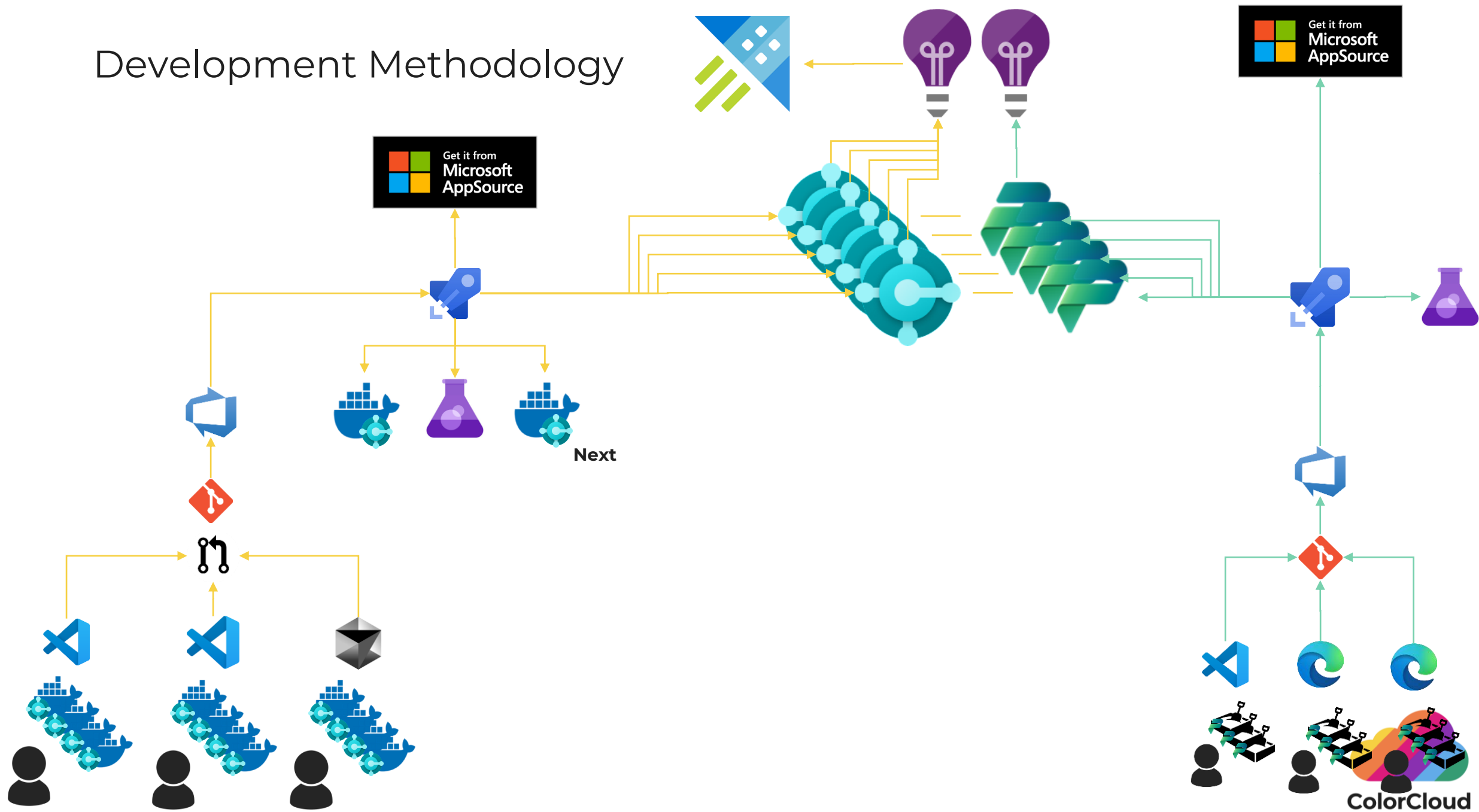
Development Methodology



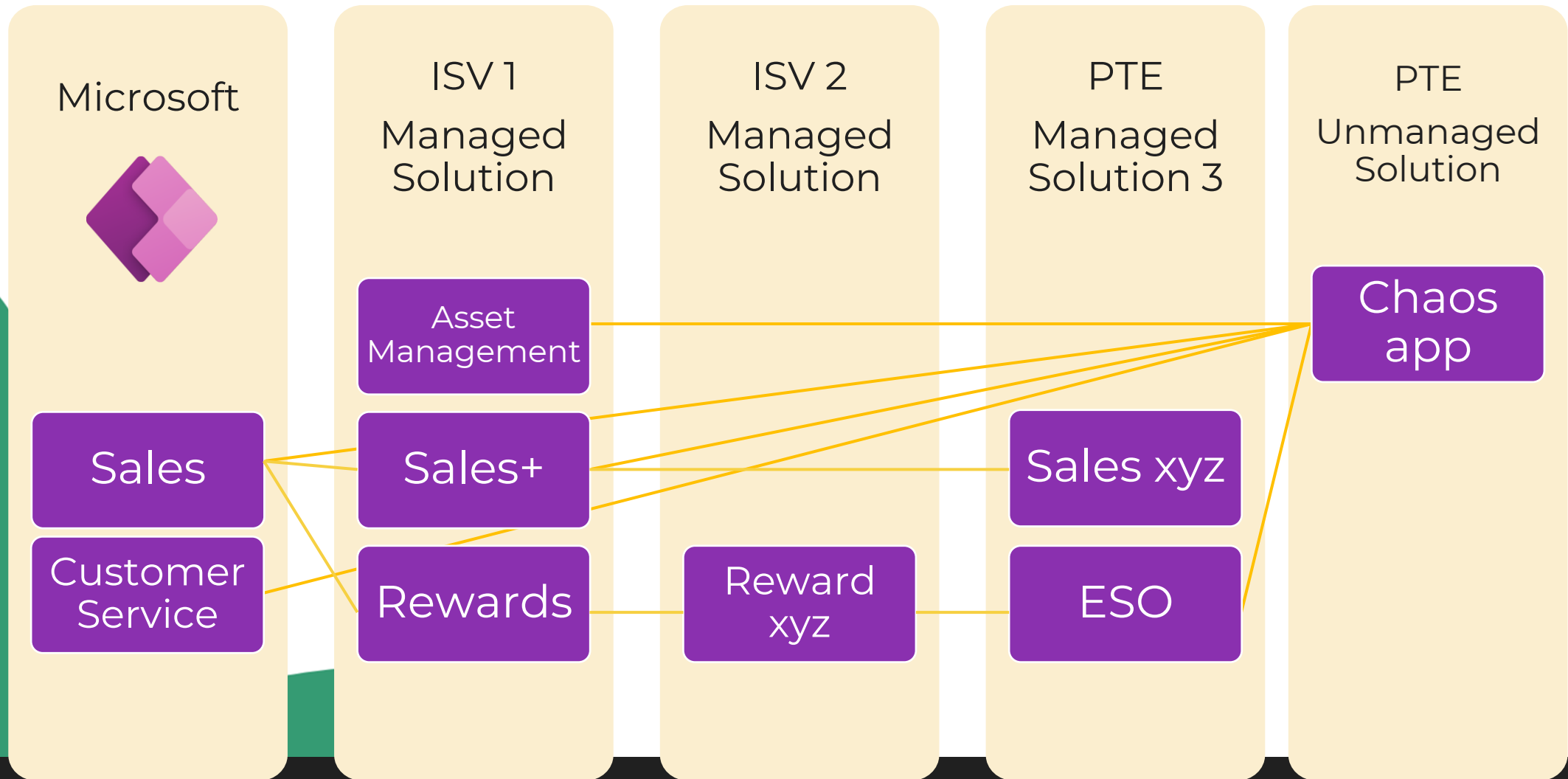
Development Methodology



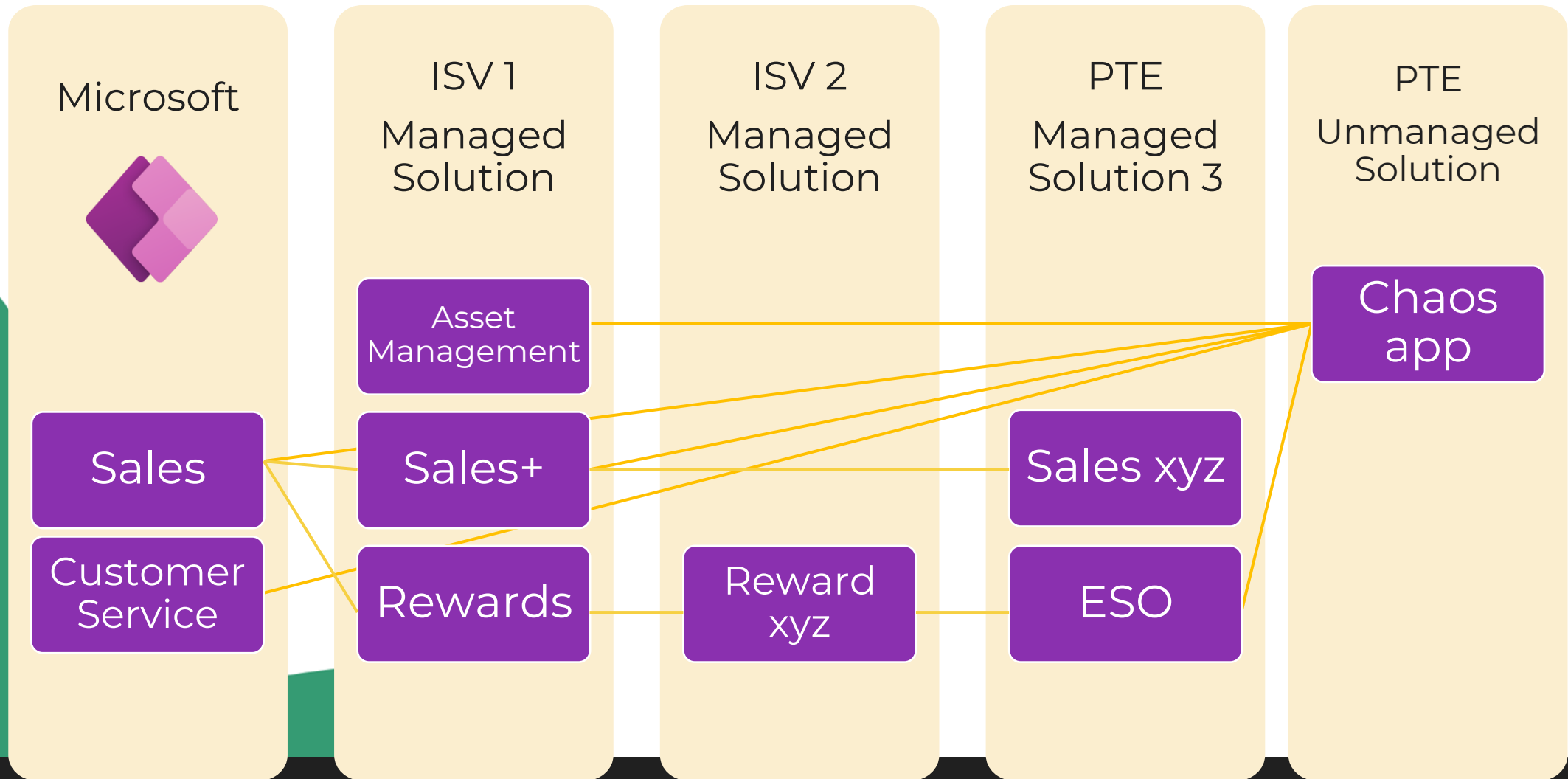
Development Methodology



Extensibility



Extensibility



Extensibility



Chaos
app



Extensibility



Canvas
app



Citizen Developer

What?

- Non-professional dev
- Typically a business user
- Typically outside traditional dev teams
- Low Code/No Code solutions



"I've built a very complex set of Power Automate flows that became the backbone of our Sales department.

When I left, it all fell apart."

Could work in Enterprise

Just doesn't work in SMB

Tine Starič



When you combine PP with BC...

Deployment

- You can't combine the installation of both apps
 - Either BC first, but what if Power Platform fails?
 - Or PP first, but what if BC fails?
- Rollback
- BC: not possible to install previous version.
- PP: IS possible to roll back



Extensibility

- **Business Central:**

- Extensibility platform

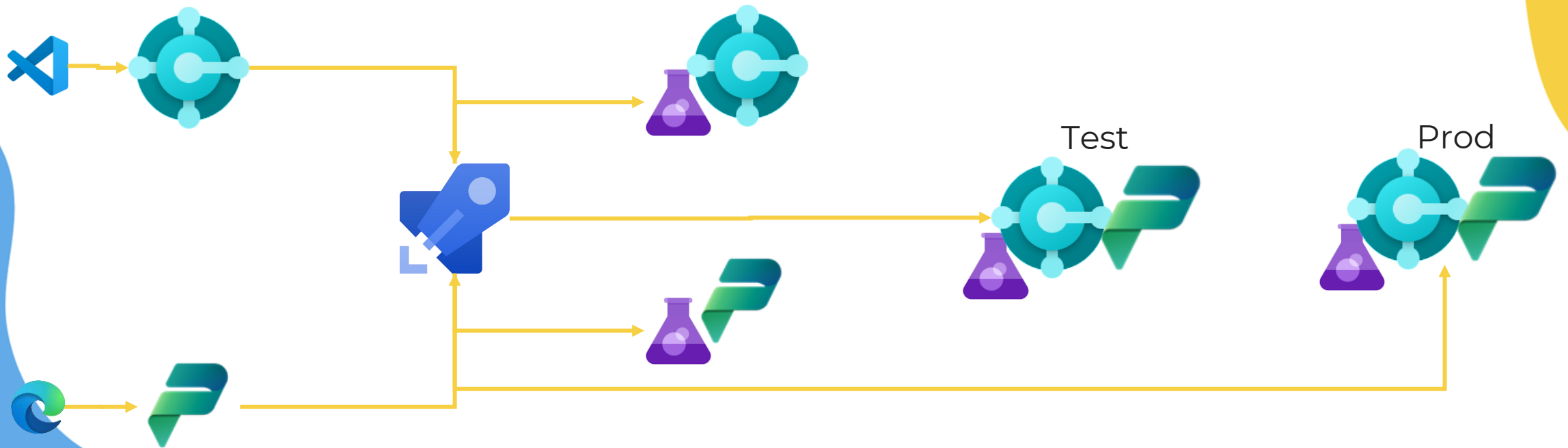
- **Power Platform:**

- PTE apps in Power Platform don't exist.
- Layer Merging does exist for Model Drive Apps.
- Canvas Apps: no extensibility.
- Conclusion: PP has an extensibility story, but not from how we use and see it.

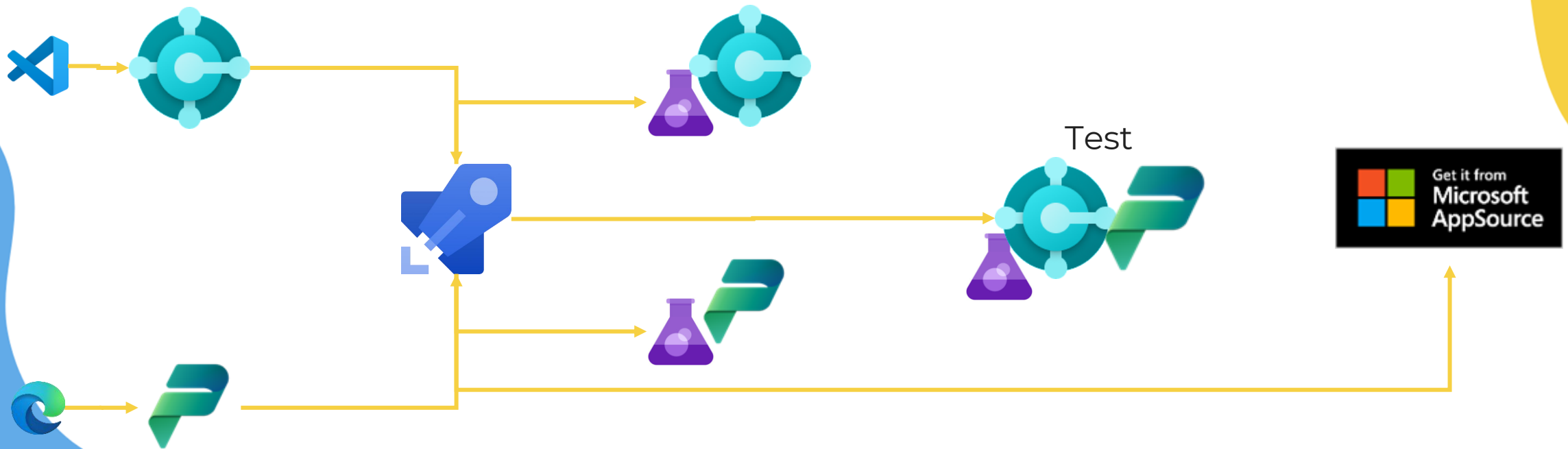
Validation

- No automated validation/tests possible.
- Now just imagine the monthly updates..

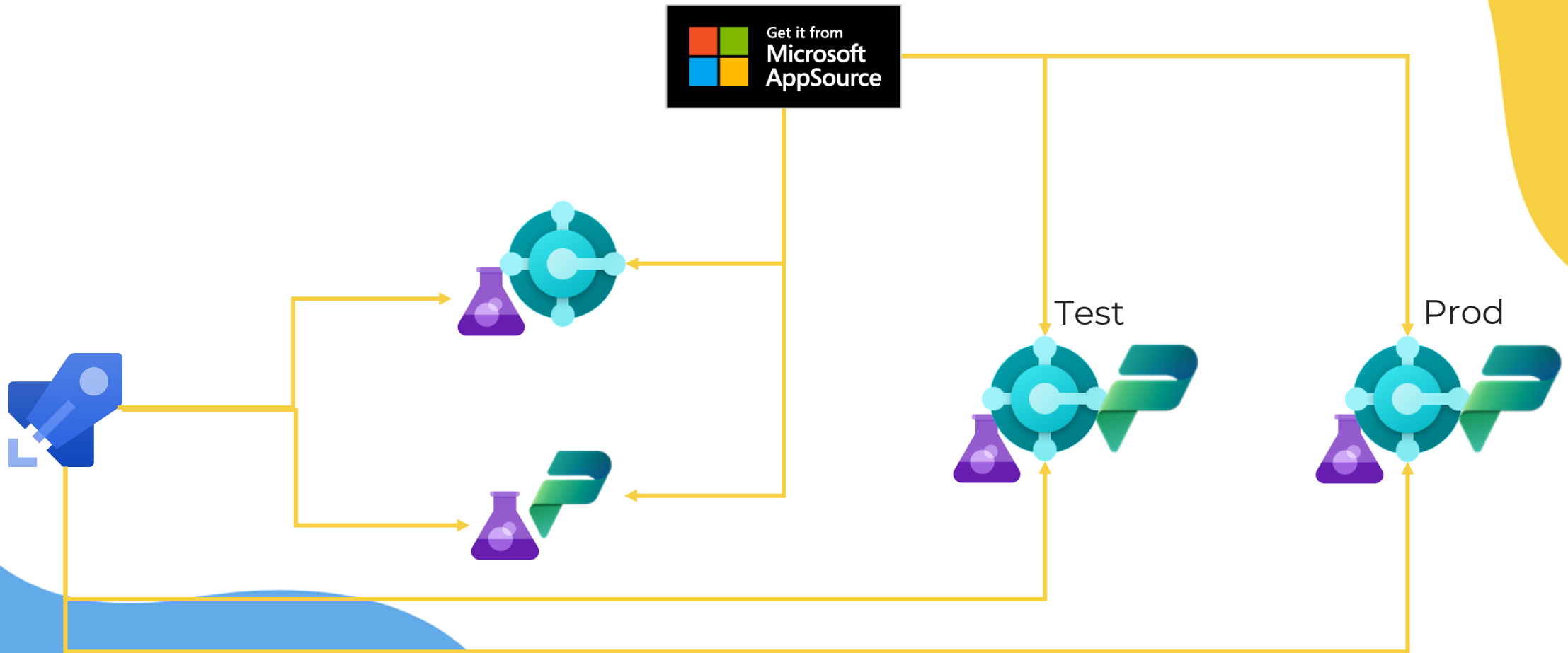
Deployment & Validation - What do we do?



Deployment & Validation - What do we do?



Deployment & Validation - What do we do?



No one should modify any system at all except through a pipeline. You cannot test that a person will do it correctly, so don't let a person do it.



Extensibility - What do we do?

- We do not clone apps
- Customizations are application features

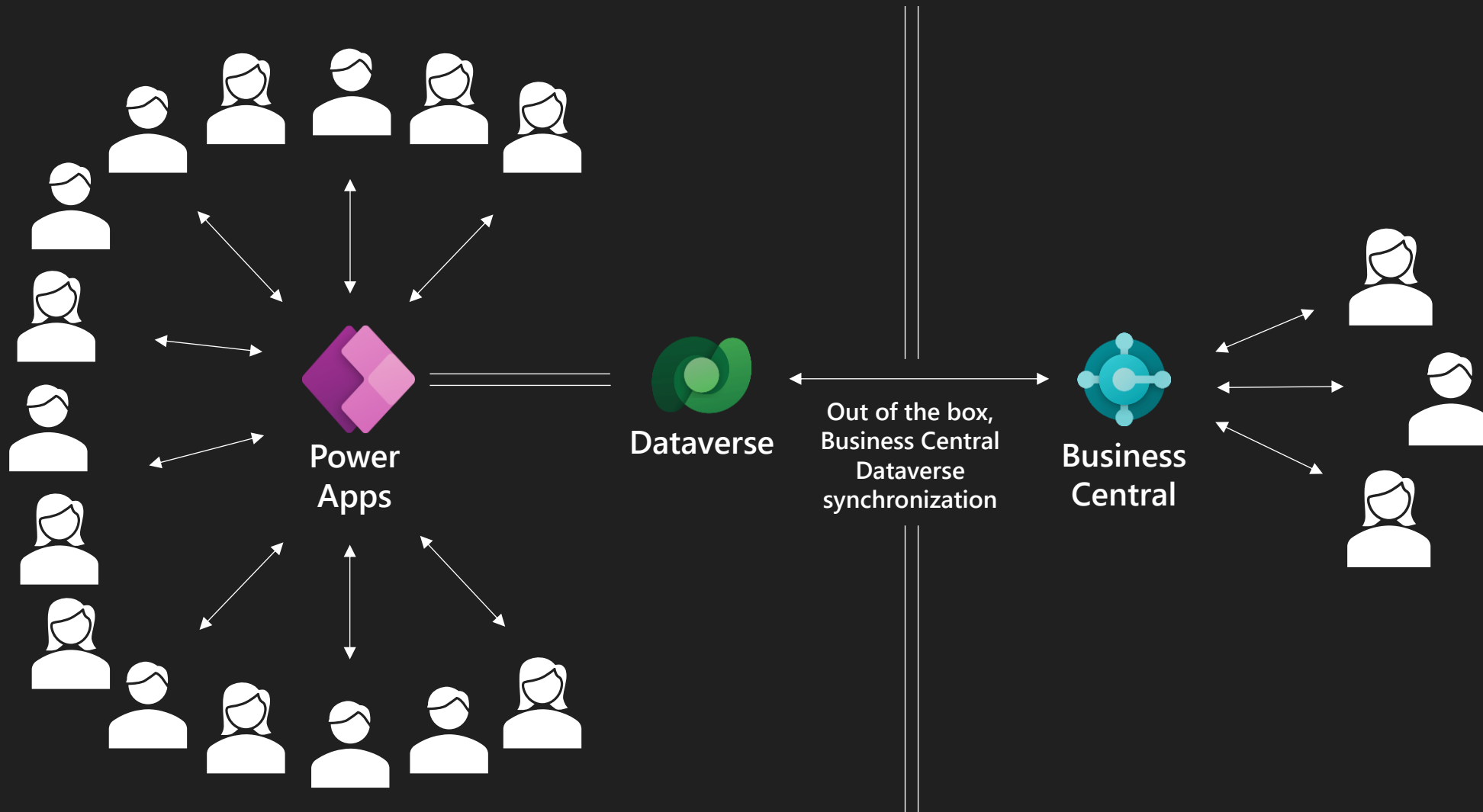


Licensing differences

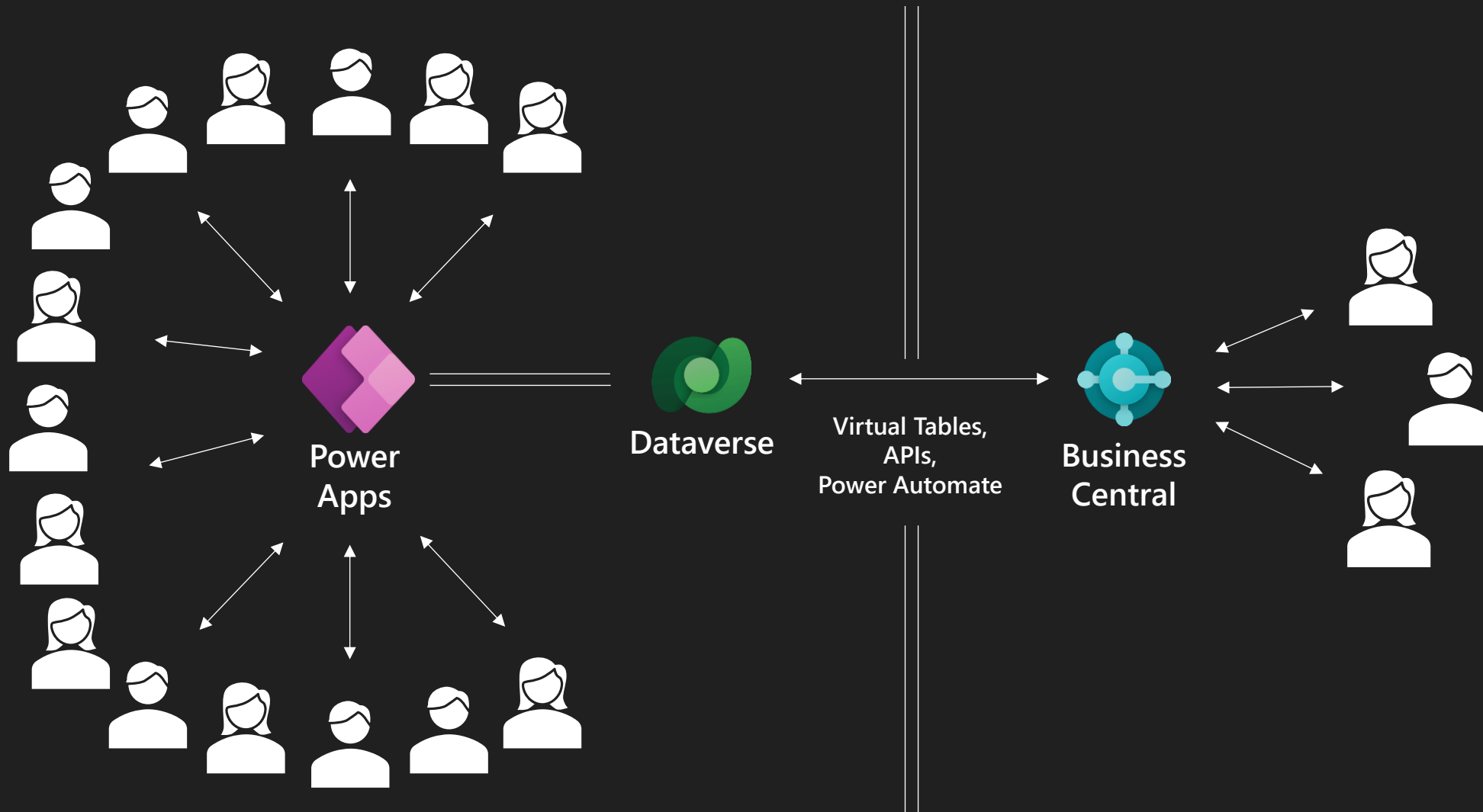
Multiplexing

The use of hardware or software to pool connections, reroute or indirectly access information, and/or reduce the number of devices or users that directly (or indirectly) access or use a product.

Multiplexing

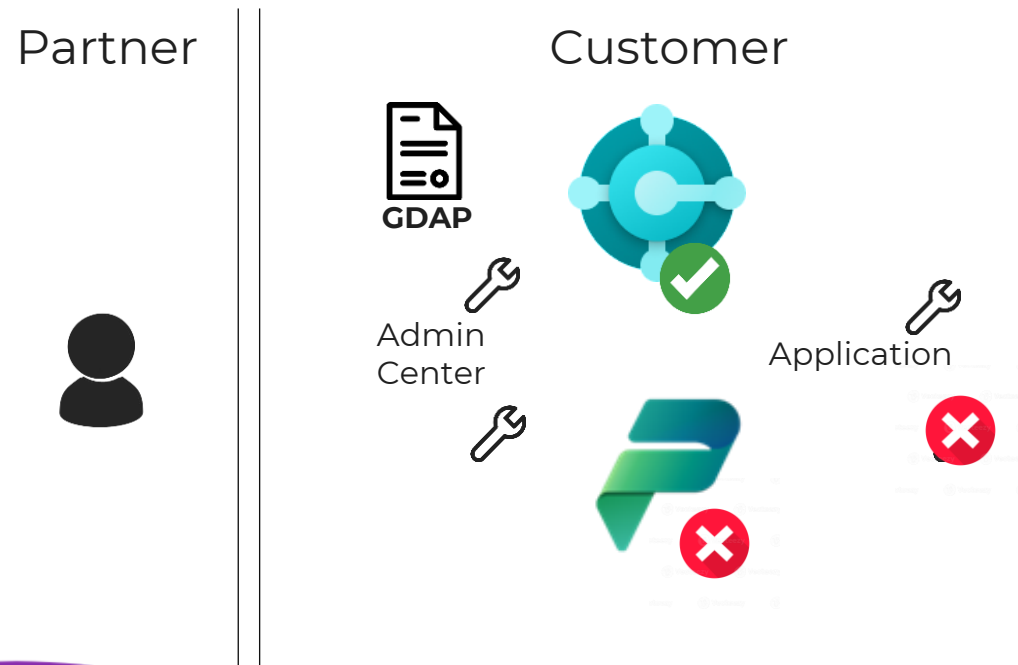


Multiplexing



Licensing differences

- **GDAP** - **G**ranular **D**elegated **A**dmin **P**ermissions



What do we do?

- For anything power platform we buy a license for any developer who needs to work on it.
- It's not a blocker, but you do want to be upfront about it.

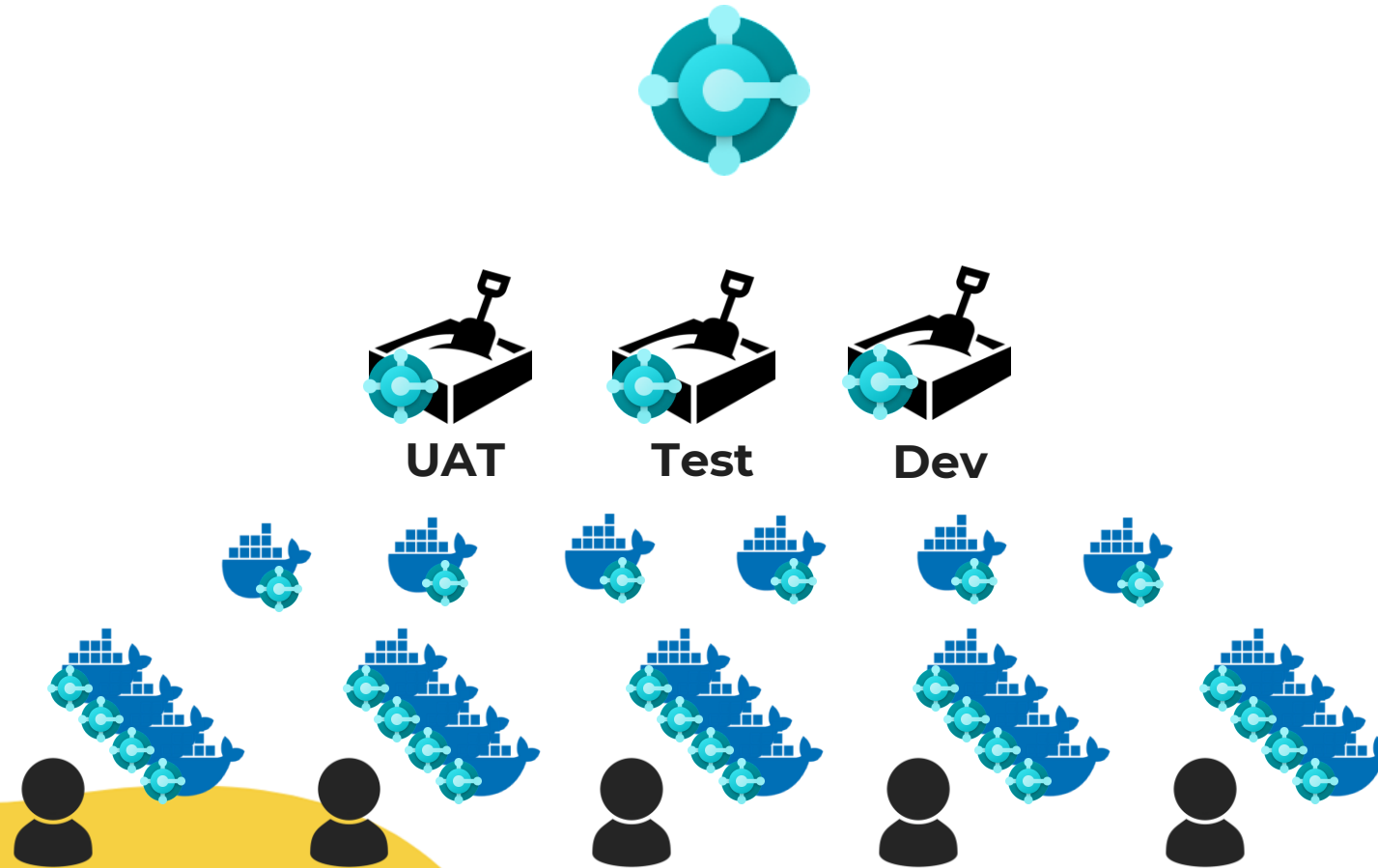
Development Environments

Things to take into account:

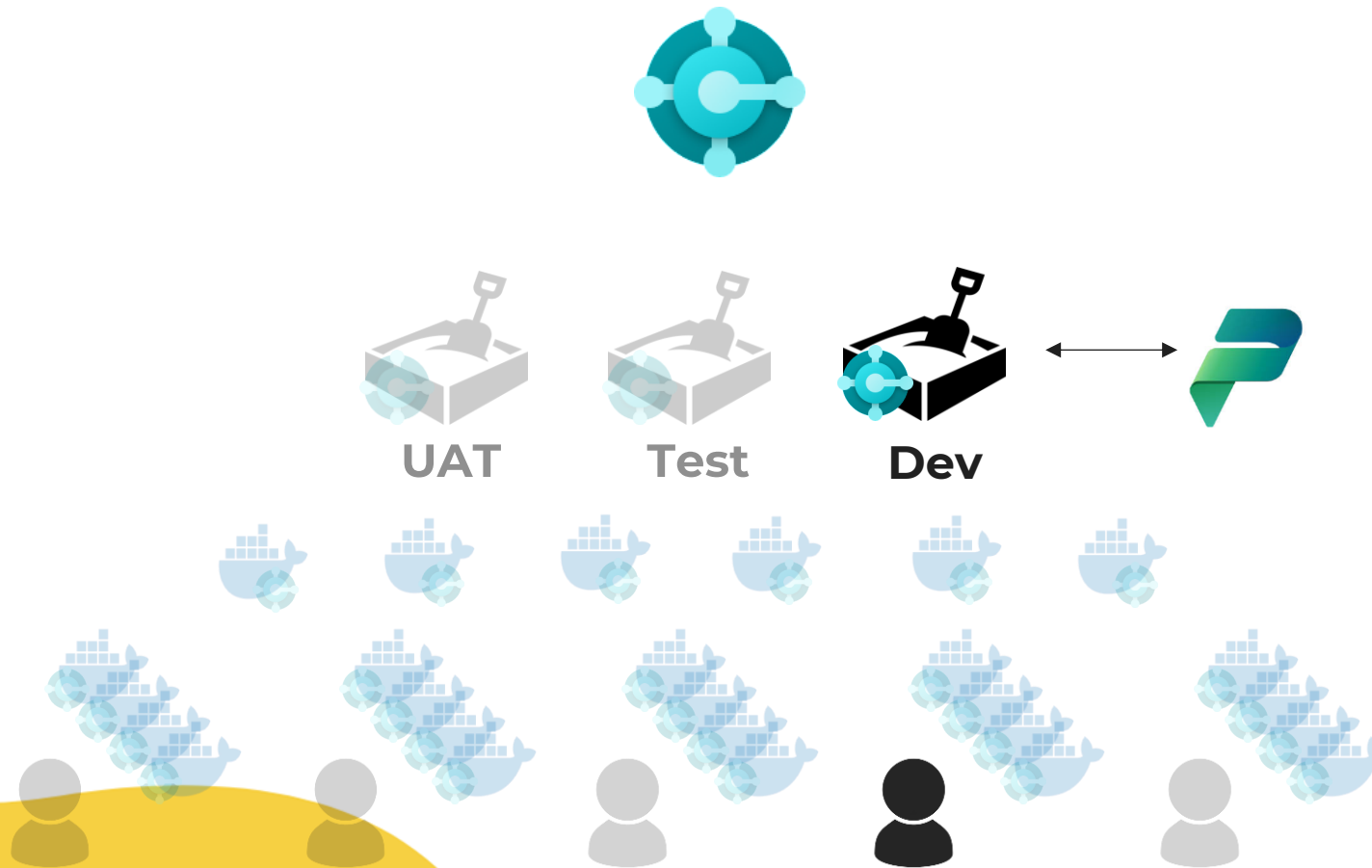
- BC only has 3 sandboxes + 1 production
- Power Platform: as many environments as you want
- BC has multiple companies in one environment, PP “expects” one

→ We can only go along with the “Greatest Common Divisor”

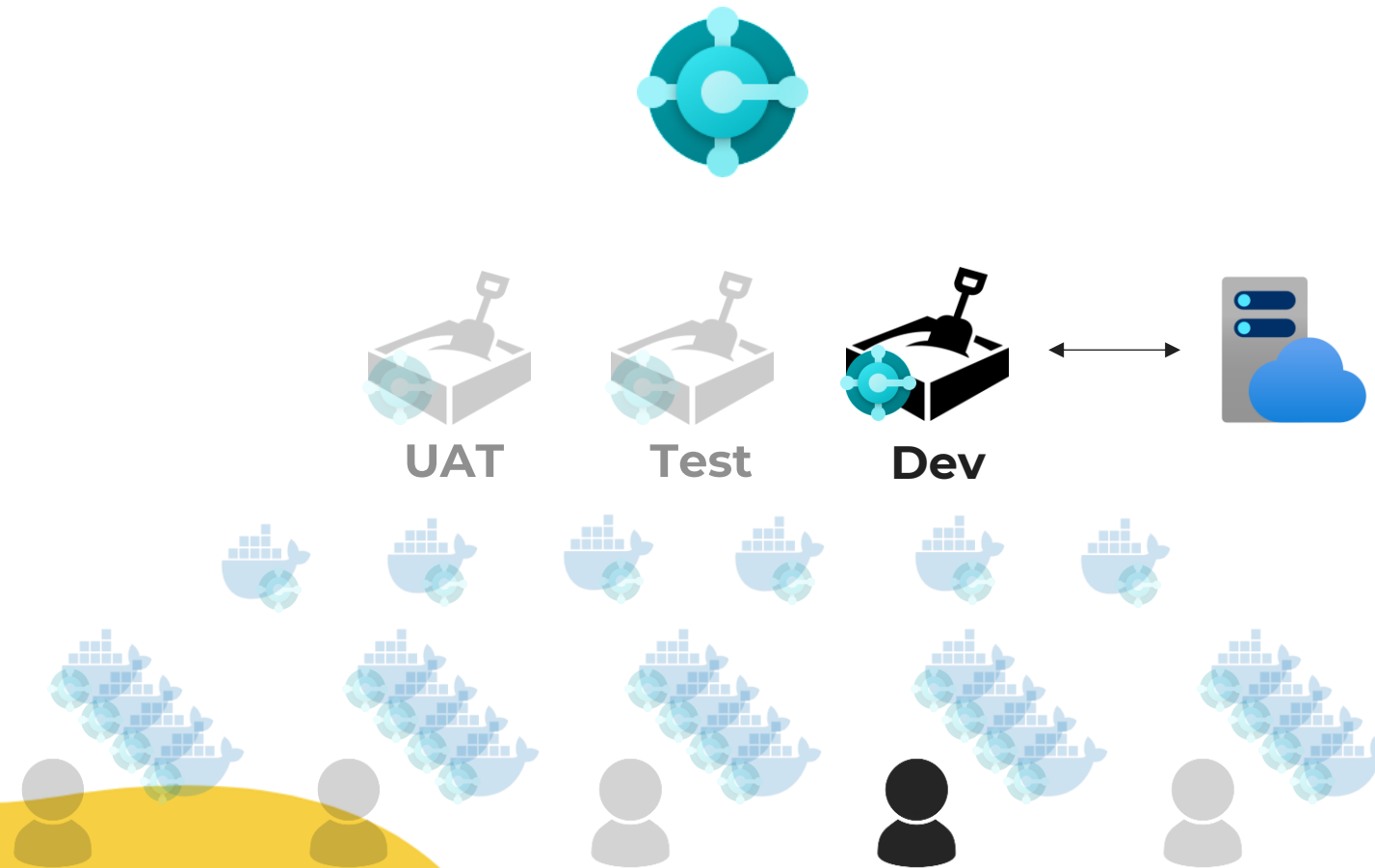
Environments - BC



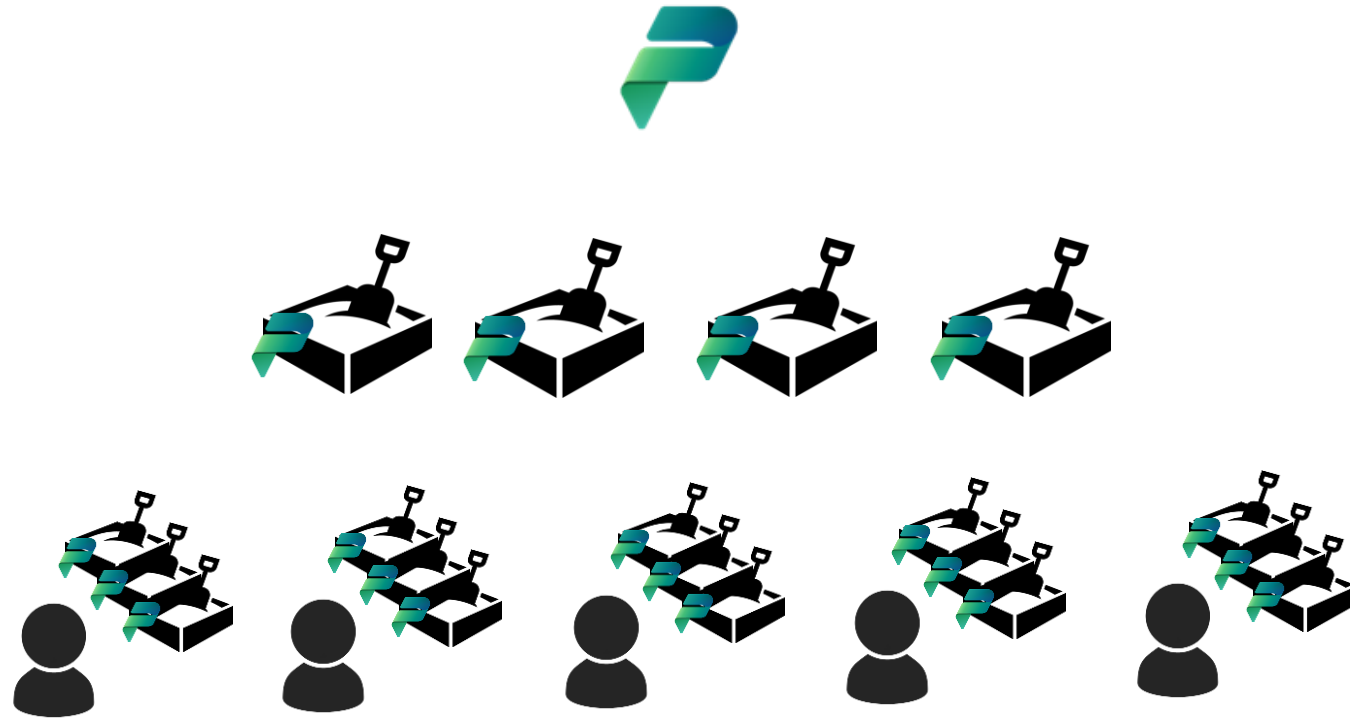
Environments - BC



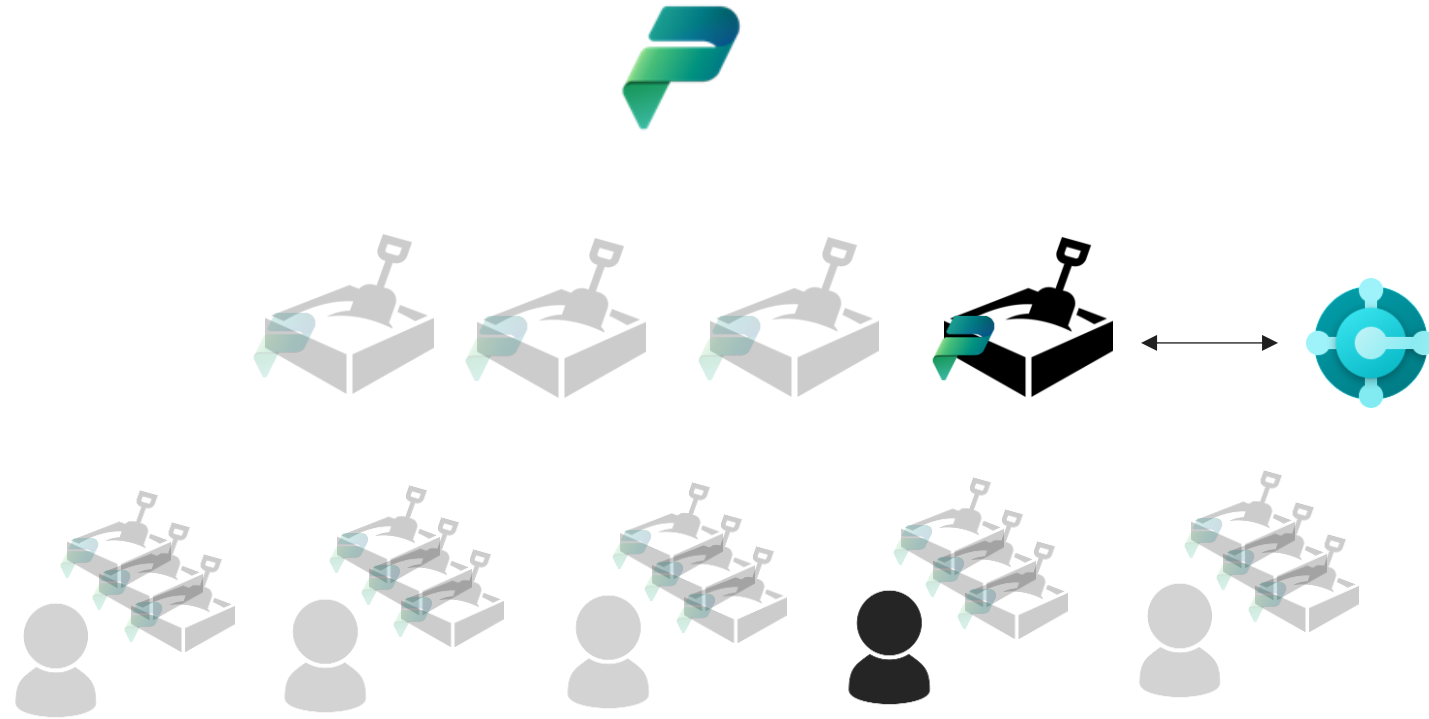
Environments - BC



Environments - PP



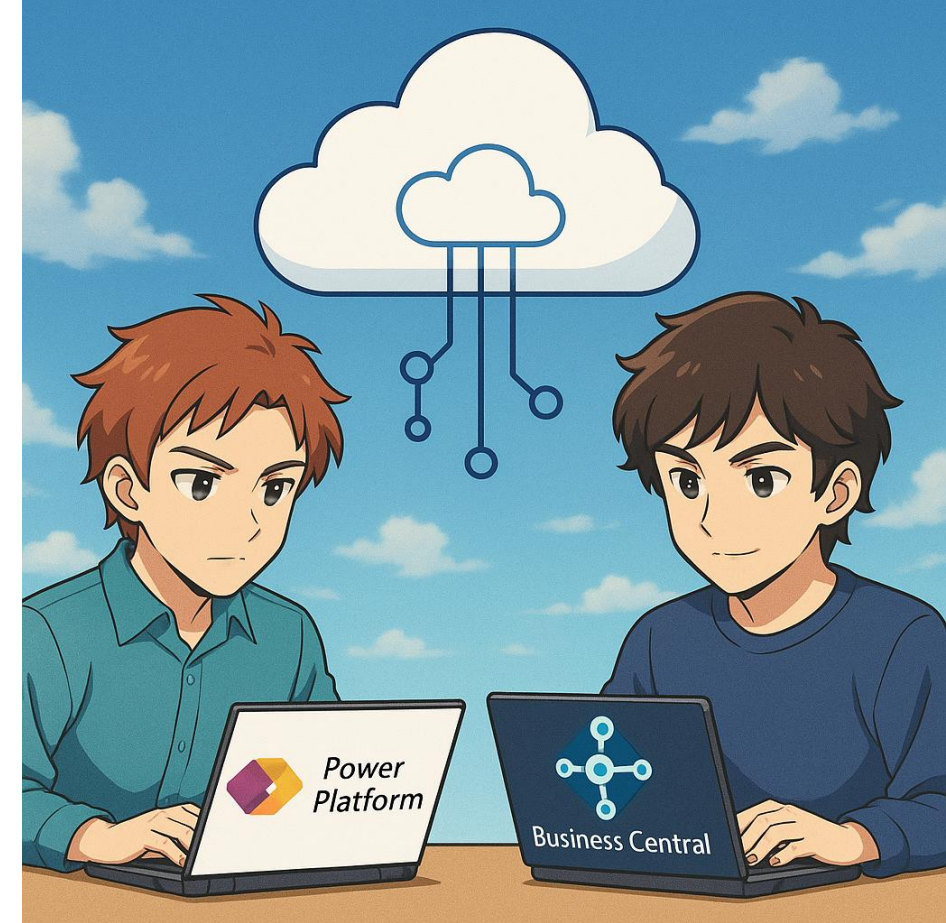
Environments - PP



What do we do?

When Data is needed on both sides:

- One developer per integration on both sides
 - in SaaS
 - Connected to one another
- Pipelines!
 - Release to QA
 - Release to LIVE
 - Release to AppSource



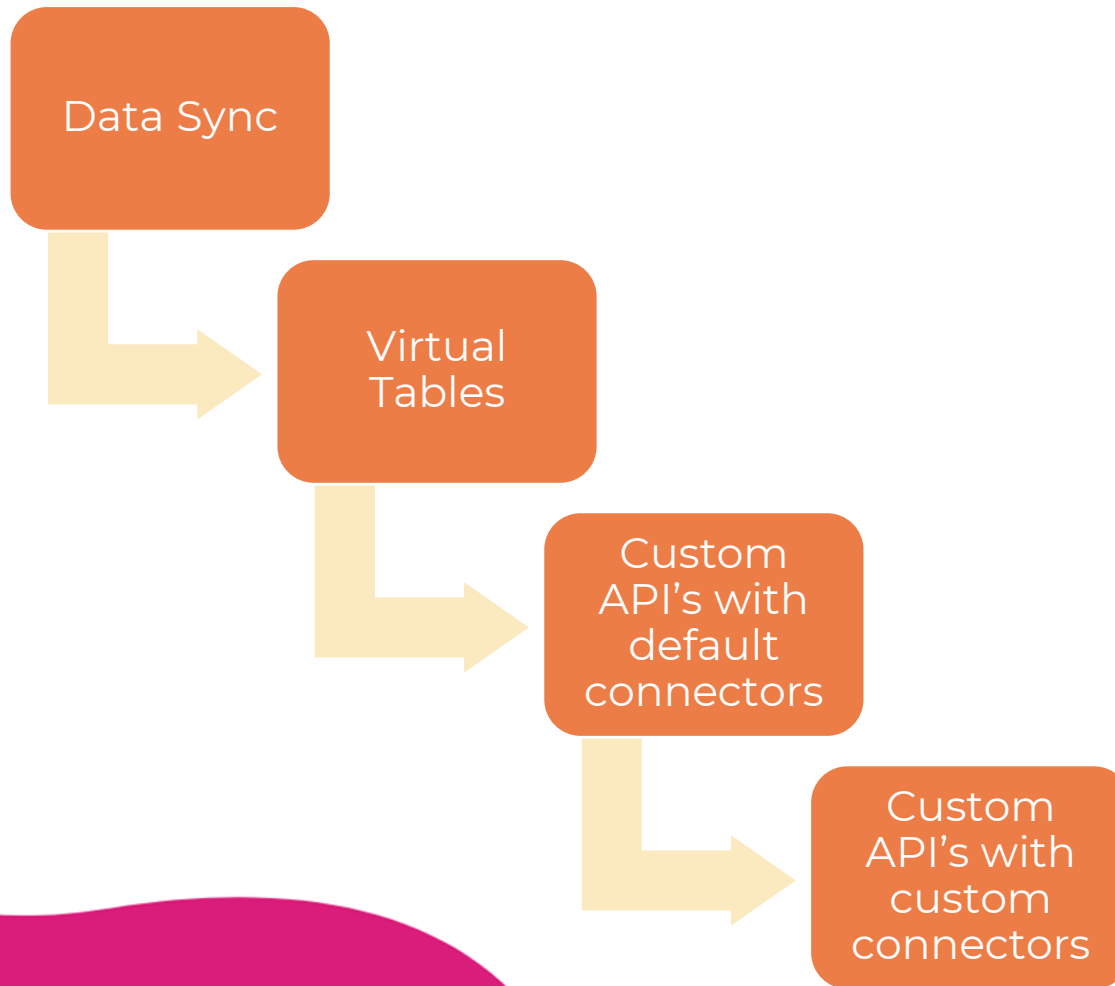
Data(verse)

- BC stores critical business data (customers, sales, inventory, ...) in an online (inaccessible) SQL Server instance.
- To work with the data in the Power Platform, takes **effort**:
 - - Use **Power Platform Connectors** to work with BC APIs
 - - **Dataverse**:
 - Data Sync
 - Virtual Tables

Dataverse & BC

- Data Sync:
 - BC copies data from BC to Dataverse and back
 - Merge Conflicts
 - Bidirectional
 - Not real time
 - Outside-the-box = development
- Virtual Tables:
 - BC API = development

What do we do



Some considerations for Dataverse

- **Avoid Bidirectional Sync in any way:** To prevent conflicts, bidirectional synchronization should generally be avoided
- **Speed of Synchronization:** Data sync speed is 10 minutes, handling up to 1,000 records per 10 minutes.

Some general considerations

- **Avoid Customizations:**

1. Default app's Business Logic
2. Integration Logic

- **Synchronous/Asynchronous Processing:** encourage async processing: most integrations don't need to be instant.

- **Support:** Define who will support the synchronization setup and ensure users in BC can see (and solve) errors/conflicts/issues.

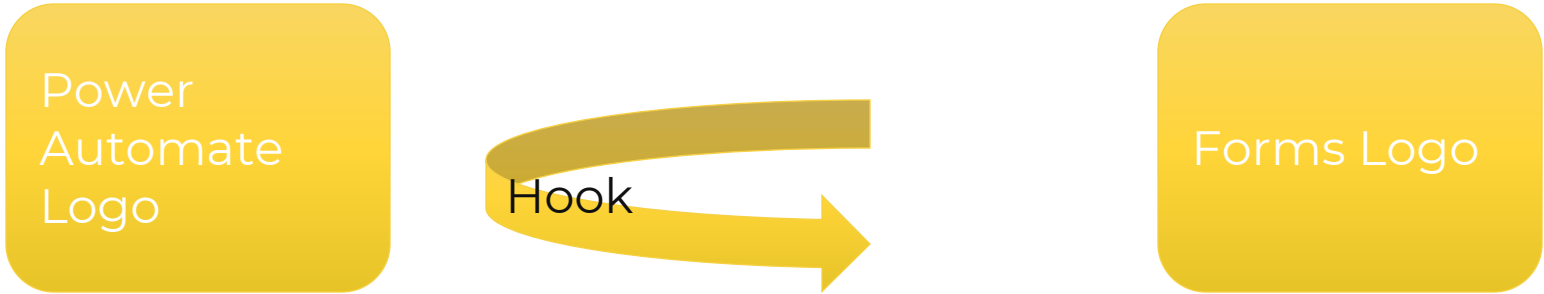
- **Customer Skill Level:** Consider the customer's skill level in managing the sync and provide additional tools if necessary

Where Power Platform really shines...

Power Automate

- Triggers are awesome
- AL cannot “hook” to webhooks natively





Power
Automate
Logo

The diagram consists of two yellow rounded squares. The left square contains the text 'Power Automate Logo' and the right square contains 'Forms Logo'. A curved yellow arrow points from the right side of the left square to the left side of the right square. The word 'Hook' is written in black text across the middle of this arrow.

Hook

Forms Logo

Power Automate

grations

```
5 references
table 50100 "Teams Setup"
{
    Caption = 'Teams Setup' page 50100 "Teams Setup"
    DataClassification = {
        PageType = Card;
        ApplicationArea = All;
        UsageCategory = Administrative;
        SourceTable = "Teams Setup";
        InsertAllowed = false;
        DeleteAllowed = false;
        Caption = 'Teams Setup';
    }
    fields
    {
        2 references
        field(1; "Primary")
        {
            Caption = 'Primary'
            DataClassification = {
                PageType = Card;
                ApplicationArea = All;
                UsageCategory = Administrative;
                SourceTable = "Teams Setup";
                InsertAllowed = false;
                DeleteAllowed = false;
                Caption = 'Teams Setup';
            }
        }
        5 references
        field(2; "Client")
        {
            Caption = 'Client'
            DataClassification = {
                PageType = Card;
                ApplicationArea = All;
                UsageCategory = Administrative;
                SourceTable = "Teams Setup";
                InsertAllowed = false;
                DeleteAllowed = false;
                Caption = 'Teams Setup';
            }
        }
        0 references
        trigger OnValueChanged
        begin
            if xRec.NewRecord
            then
                SetField(Enabled, true);
            end;
        end;
    }
    layout
    {
        area(Content)
        {
            group(General)
            {
                Caption = 'General'
                field(Enabled; "Enabled")
                {
                    ApplicationArea = All;
                    ToolTip = 'Specifies whether the team is enabled.'
                }
            }
            group(Authentication)
            {
                Caption = 'Authentication'
                field(Enabled; "Enabled")
                {
                    ApplicationArea = All;
                    ToolTip = 'Specifies whether the team is enabled.'
                }
            }
        }
    }
}

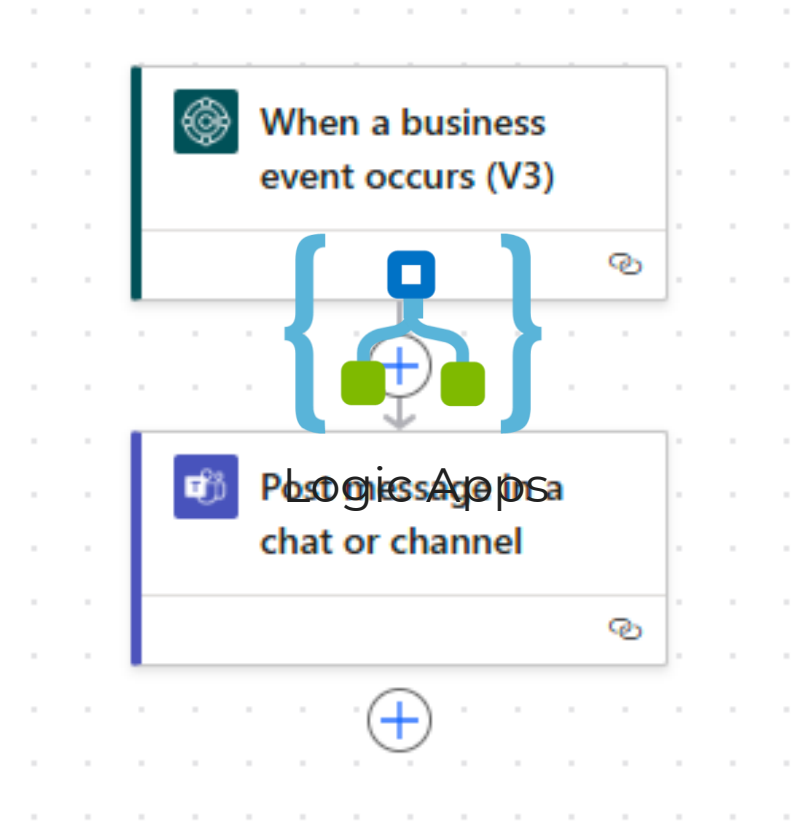
1 reference | 0% Coverage
local procedure GetAccessToken() Success: Boolean
var
    RequestUrl: Text;
    ResponseText: Text;
    FormContent: Text;
    FormContent2: Text;
    JsonObject: Text;
    ClientId: Text;
    ClientSecret: Text;
    TenantId: Text;
begin
    if not TeamId
    then
        Error('Team ID is required.')
    end;
    RequestUrl := 'https://login.microsoftonline.com/' + TenantId + '/oauth2/token?client_id=' + ClientId + '&grant_type=client_credentials';
    ResponseText := GetHttpData(RequestUrl, 'POST', ClientSecret);
    if ResponseText = ''
    then
        Error('Failed to get access token.')
    end;
    JsonObject := ParseJson(ResponseText);
    ClientId := JsonObject['access_token'];
end;

2 references | 0% Coverage
procedure SendChannelMessage(ChannelId: Text; MessageText: Text)
var
    RequestBody: Text;
begin
    ResponseText := '';
    // Create request
    HttpRequest.Method(Method);
    HttpRequest.SetRequestUri(Uri);
    HttpRequest.GetHeaders(HttpHeaders);
    // Format the request body
    RequestBody := Format(MessageText);
    // Set headers
    HttpHeaders.Remove('Content-Type');
    HttpHeaders.Remove('Accept');
    HttpHeaders.Add('Content-Type', ContentTypeJson);
    HttpHeaders.Add('Accept', AcceptJson);
    HttpHeaders.Remove('Authorization');
    HttpHeaders.Add('Authorization', StrSubstNo('Bearer %1', ClientId));
    // Set body if provided
    if RequestBody <> ''
    then
        HttpRequest.SetRequestBody(RequestBody);
    end;
    SendRequest(HttpRequest);
end;
```

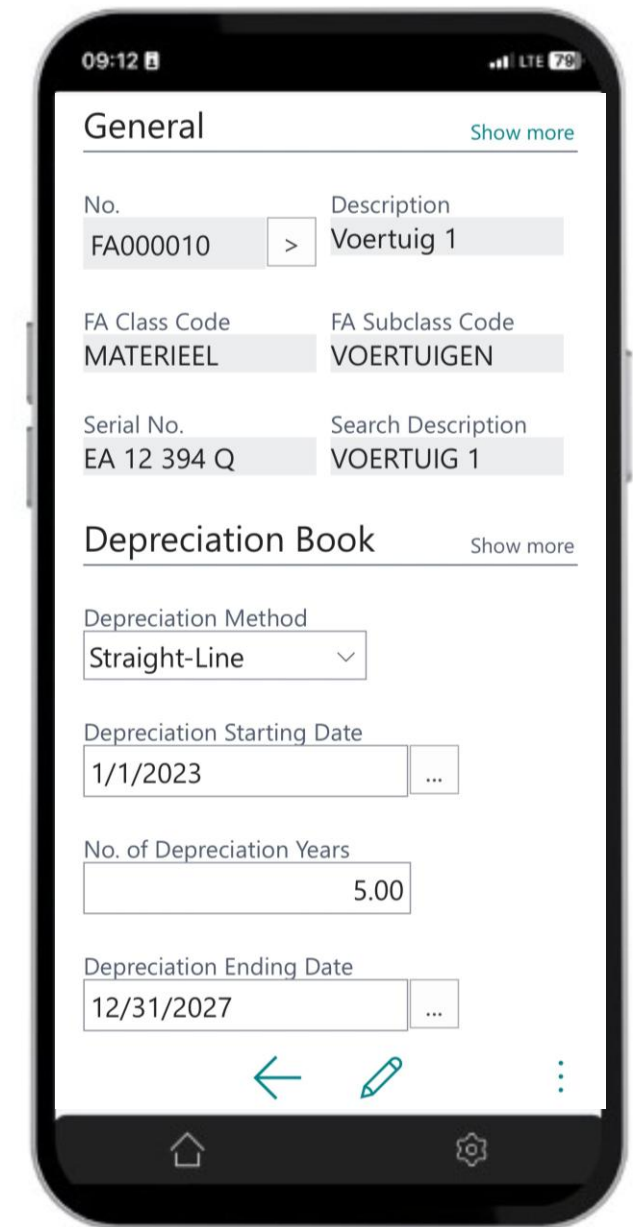
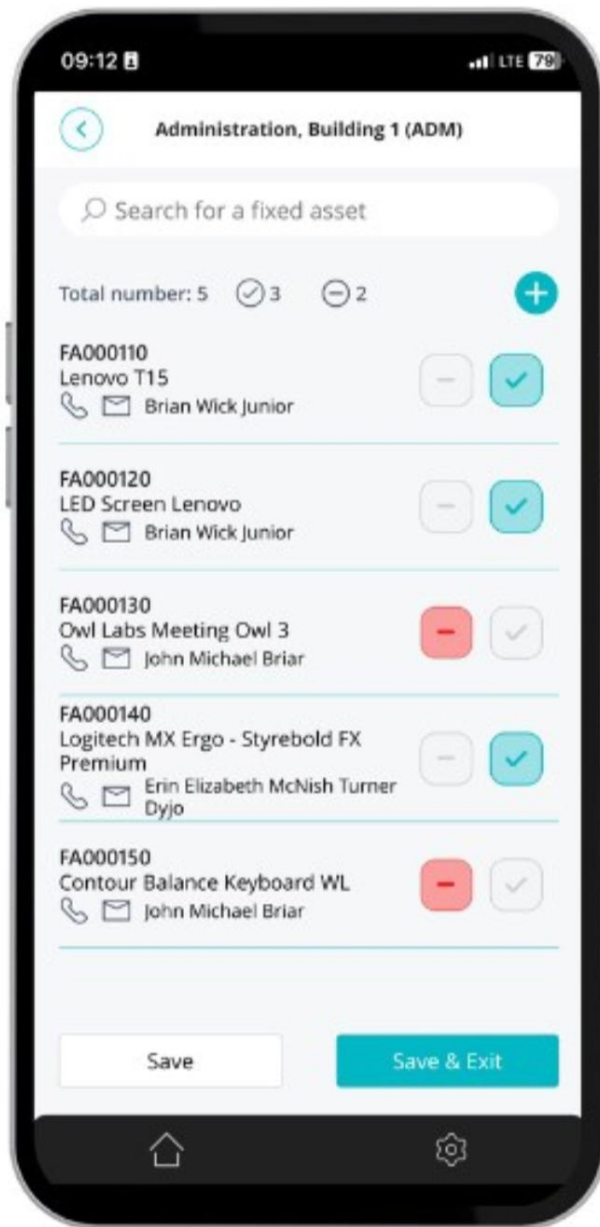


Power Automate

- Simplify non-critical integrations
- It's Power Automate not Power Integrate 🤨



Simplistic UI



Offline Mode



09:12 LTE 79%

Administration, Building 1 (ADM)

Search for a fixed asset

Total number: 5 ✓ 3 ✗ 2 +

FA000110	Lenovo T15	-	✓
📞 📧 Brian Wick Junior			
FA000120	LED Screen Lenovo	-	✓
📞 📧 Brian Wick Junior			
FA000130	Owl Labs Meeting Owl 3	-	✓
📞 📧 John Michael Briar			
FA000140	Logitech MX Ergo - Styrebold FX Premium	-	✓
📞 📧 Erin Elizabeth McNish Turner Dyjo			
FA000150	Contour Balance Keyboard WL	-	✓
📞 📧 John Michael Briar			

Save Save & Exit

09:12 LTE 79%

General Show more

No.	Description
FA000010	Voertuig 1

FA Class Code	FA Subclass Code
MATERIEEL	VOERTUIGEN

Serial No.	Search Description
EA 12 394 Q	VOERTUIG 1

Depreciation Book Show more

Depreciation Method

Straight-Line ▼

Depreciation Starting Date

1/1/2023 ...

No. of Depreciation Years

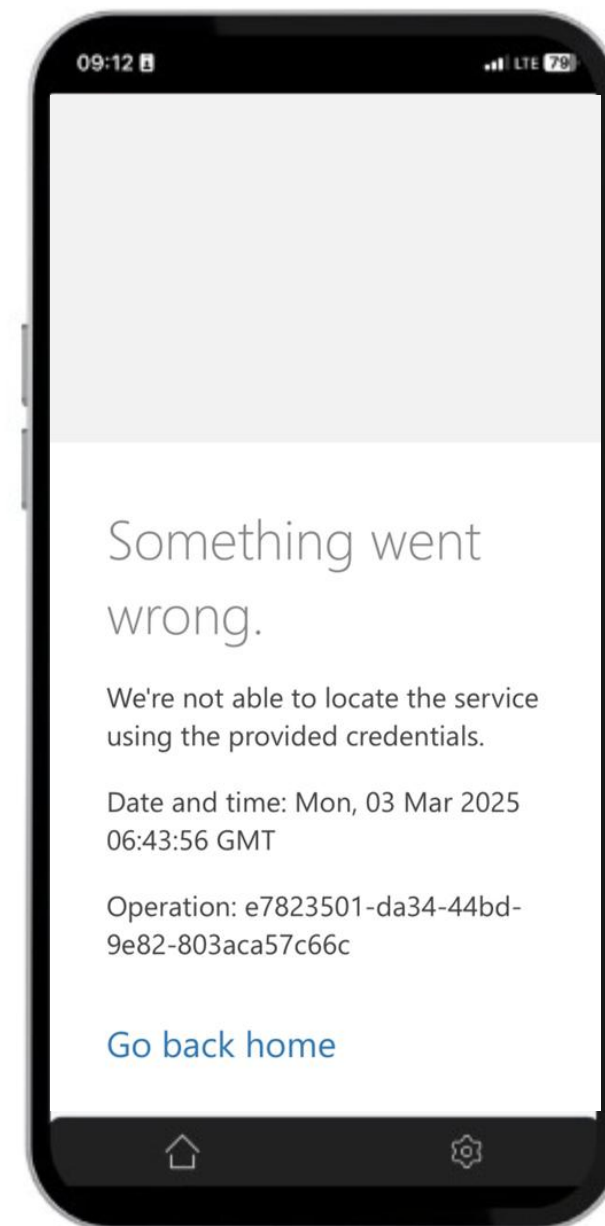
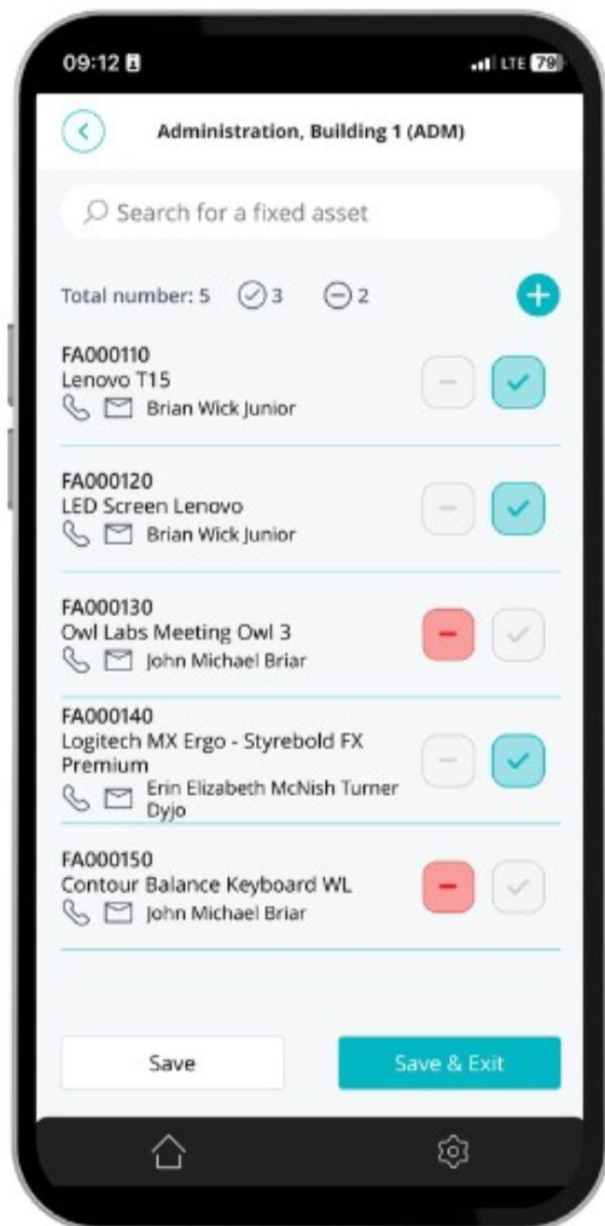
5.00

Depreciation Ending Date

12/31/2027 ...

← ✎ ⋮

Offline Mode









Power Pages



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Item Catalog

See a list of all our wonderful items that will make you

Product ID ↑	Name	Description
1896-S	ATHENS Desk	
1900-S	PARIS Guest Chair, black	
1906-S	ATHENS Mobile Pedestal	
1908-S	LONDON Swivel Chair, blue	
1920-S	ANTWERP Conference Table	10.00 USD420.40
1925-W	Conference Bundle 1-6	0.00 USD122.50
1928-S	AMSTERDAM Lamp	8.00 USD35.60



ATHENS Desk

The only option for truly sky-rocketing your productivity. The perfect addition to your home office.



PARIS Guest Chair

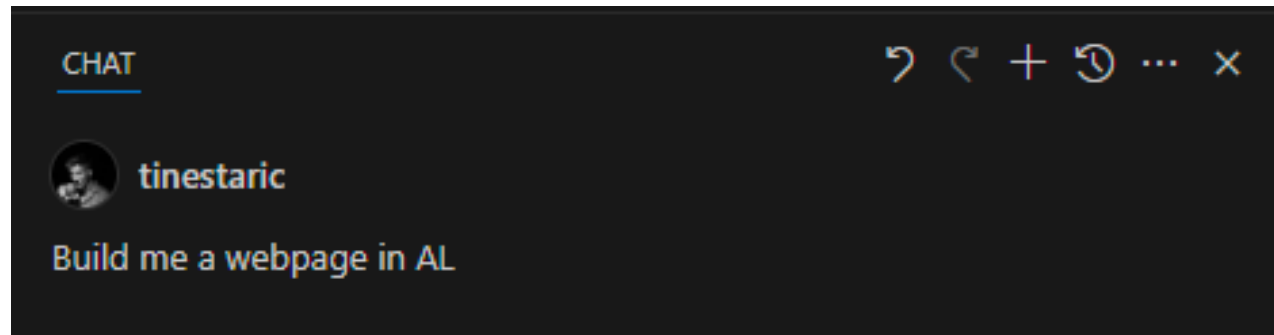
Seating experience reimagined. You will never want to get up ever again.



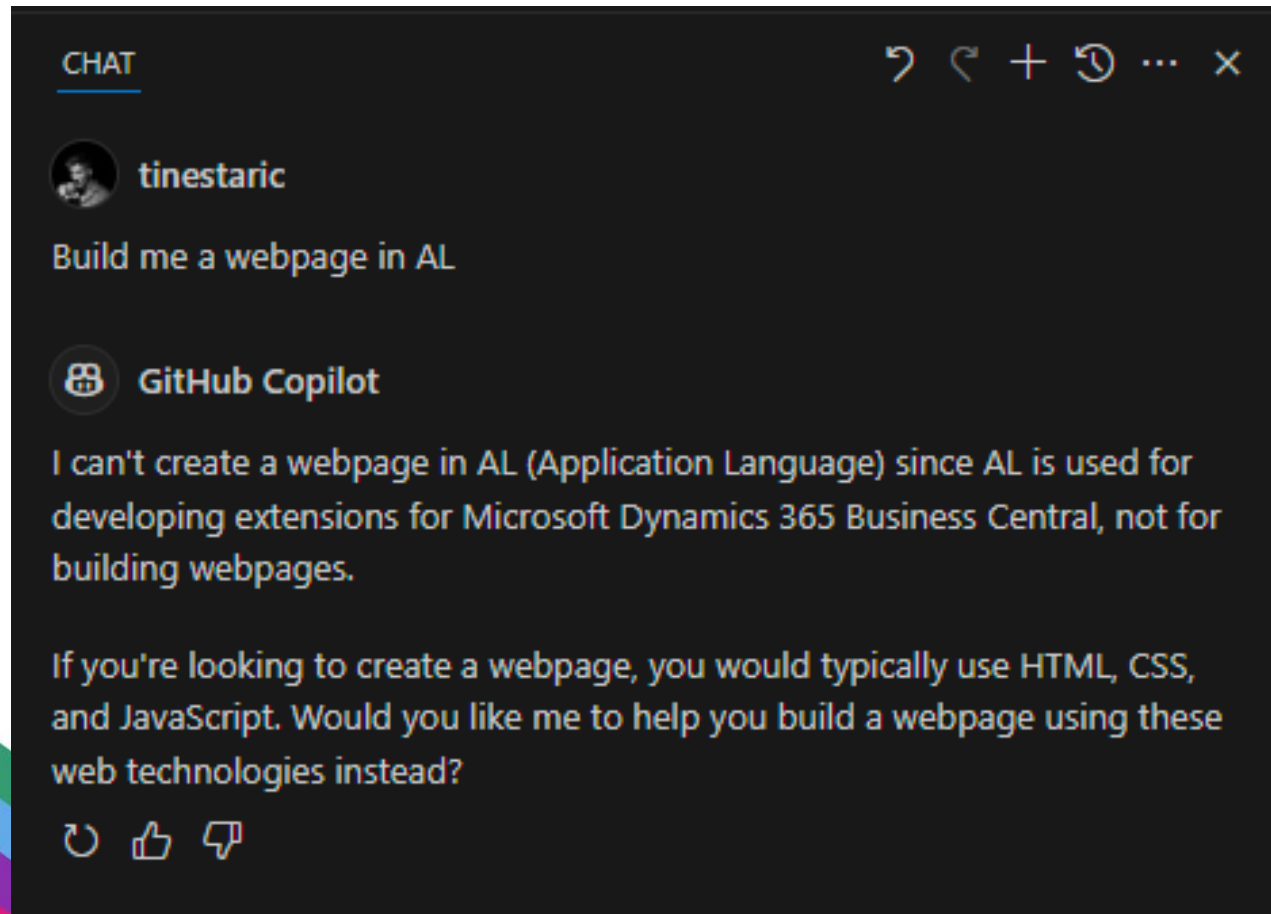
ATHENS Mobile Pedestal

The perfect addition to any balcony. Give the people an option to put that perfect cup of coffee down for a second and enjoy the morning air.

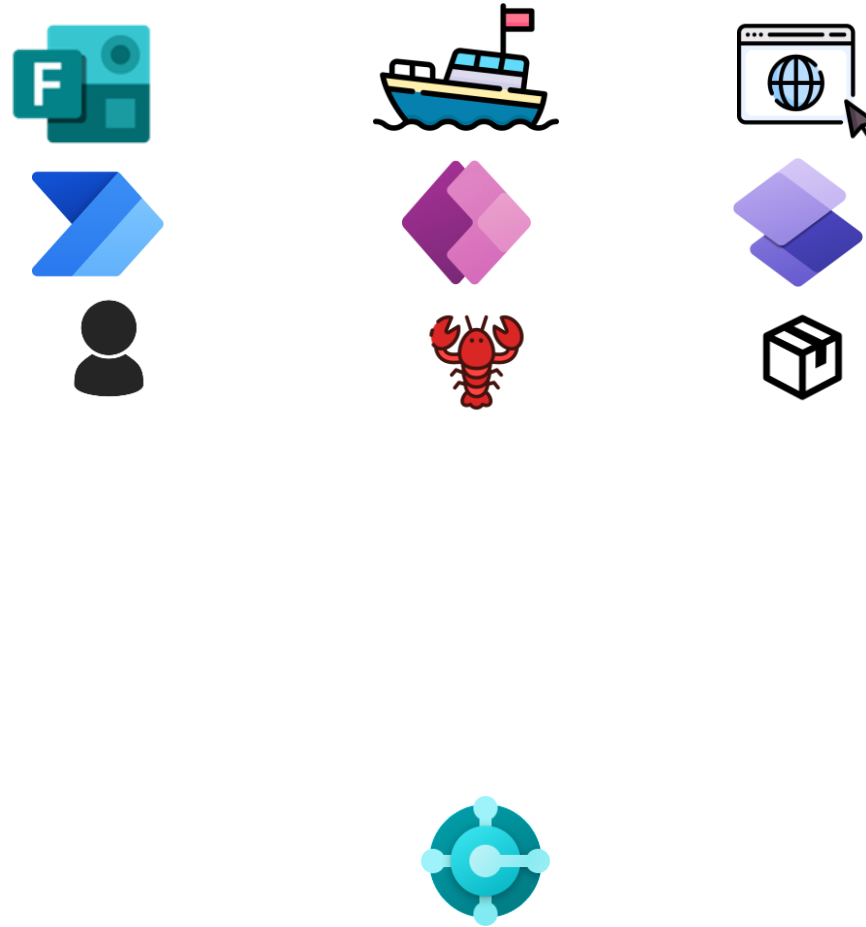
Power Pages



Power Pages



Data collection is key



Model-Driven Apps

- We honestly don't have a good example. It was always better to build it in BC
- Unless you're supporting a process that has nothing to do with BC

Key takeaways

Don't bring a low-code cannon to an AI knife fight.

Business Central already covers much of what Power Platform offers—so use it where it truly adds value.

You don't "*just do Power Platform*" without
making a mess



**You don't “*just do Business Central*” without making
a mess**

A horizontal bar with a rainbow gradient, transitioning from yellow on the left to orange, red, pink, purple, blue, and green on the right.

How do we do what we do?

We implement **pipelines** to ensure smooth and efficient processes.

We **buy licenses** for every developer to maintain compliance and functionality.

We **plan developers** accordingly on both sides to ensure seamless integration.

We do not skip **applying best practices**

+ We motivate customers to use best practices



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THANK YOU!



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Q&A



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Feedback

Please rate my session to help us improve!

Take part in ColorCloud Raffle 🔥

Every submitted feedback counts as one raffle ticket! **Good luck** 🍀

